

SUPPLEMENTARY MATERIAL

S1 Examples of benefits of $\hat{S}^{ei}$: Illustrations in four examples of sparsity and informativeness of the blocks of the benefits of $\hat{S}^{ei}$ in oracle selection properties.

S2 Complementary material on numerical illustrations: Reports of false discovery rates, power, mean squared error and computation time in the numerical illustrations of the performance of $\hat{S}^{ei}$ and Tran-s-ell0.

S3 Proofs

S1 Examples of benefits of $\hat{S}^{ei}$

We illustrate here the benefits of $\hat{S}^{ei}$ over $\hat{S}^{sd}$ in four concrete examples contemplating different scenarios of sparsity regimes and informativeness of the blocks, summarized in Table S1. We assume an orthonormal design such that $\rho(\mathbf{X}) = 1$ for simplicity and focus on a sparse setting where $\ln(s) = o(\ln(p - s))$, with two blocks ($b = 2$). The smallest non-zero $|\beta_i^*|$, $\beta_{\min}^*$, is assumed to be located in block 2, without loss of generality.

Table S1: Sparsity and block assumptions in Scenarios 1–4

Scenario	$p - s$	$p_1 - s_1$	$p_2 - s_2$	s_1	s_2
1	$3n/2$	$3n/2 - \sqrt{n}$	$\sqrt{n}$	$3\ln(n)/2$	$3\ln(n)/2$
2	$\exp(n/16)$	$\exp(n/16) - n^2$	n^2	$3\ln(n)/2$	$3\ln(n)/2$
3	n	$n - \ln(n)$	$\ln(n)$	$3\ln(n)/2$	$3\ln(n)/2$
4	n	$n/2$	$n/2$	$3\ln(n)/2$	$3\ln(n)/2$

In Scenarios 1 to 3, blocks are informative as block B_1 is sparser than block B_2 . In Scenario 1 the blocks are moderately informative, in that $[s_2/(p_2 - s_2)]/[s_1/(p_1 - s_1)] = O(\sqrt{n})$. In Scenario 2 they are highly discriminative, in that $[s_2/(p_2 - s_2)]/[s_1/(p_1 - s_1)] = O(\exp(n/16))$. In Scenario 3, they are highly discriminative in that β^* is sparse overall, but it is non-sparse within block 2 ($s_2 > p_2 - s_2$). In Scenario 4, the block partition is non-informative of the distribution of inactive parameters: each block has half of them.

Recall that the right-hand sides in (17) and (18) are the smallest signal strength allowed by the sufficient assumptions for consistency (A1–A2) with $\hat{S}^{ei}$ and $\hat{S}^{sd}$ respectively. Denote them $\tilde{\beta}_{\min,j}^*$ and $\tilde{\beta}_{\min}^*$ respectively, that is, for the informed selector $\hat{S}^{ei}$,

$$\tilde{\beta}_{\min,j}^* = \frac{\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}}{\sqrt{\rho(\mathbf{X})n}} \quad \text{for each } j = 1, \dots, b,$$

and for the standard $\hat{S}^{sd}$,

$$\tilde{\beta}_{\min}^* = \frac{\sqrt{\ln(p - s)} + \sqrt{\ln(s)}}{\sqrt{\rho(\mathbf{X})n}}.$$

Figure S1 plots $\tilde{\beta}_{\min,2}^*$ and $\tilde{\beta}_{\min}^*$ in Scenarios 1–4. The smallest signal strength allowed for $\hat{S}^{ei}$ in block 2 is smaller than that allowed for $\hat{S}^{sd}$, with the magnitude of the difference

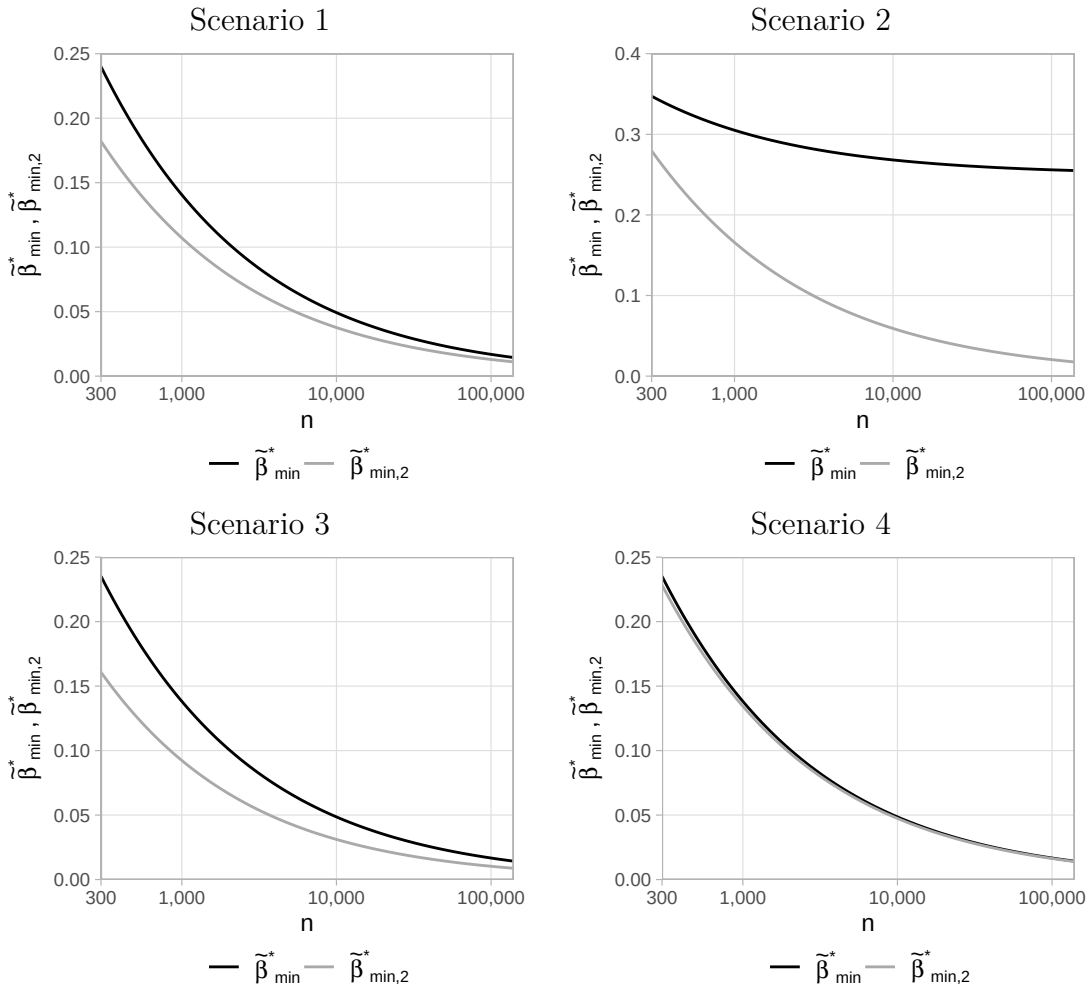


Figure S1: Smallest signal strength $\tilde{\beta}_{\min,2}^*$ and $\tilde{\beta}_{\min}^*$ in Scenario 1 (top left), 2 (top right), 3 (bottom left), and 4 (bottom right) as in Table S1 (Section S1)

depending on the informativeness of the block partition. It is the largest in the highly informative Scenario 2 and essentially null in the non-informative Scenario 4. Scenario 2 also provides a straightforward example where incorporating external information makes possible consistent recovery. For any constant $\beta_{\min}^* = \beta_{\min,2}^* < 0.25$, consistent recovery is impossible with $\hat{S}^{sd}$, but possible with $\hat{S}^{ei}$.

We compare next the convergence rates associated with $\hat{S}^{ei}$ and $\hat{S}^{sd}$ for penalties chosen by an oracle, that is, OR^{ei} and OR^{sd} respectively (see Theorem 3.4). Figure S2 plots the ratio OR^{ei}/OR^{sd} as given in (19) in Scenario 1 to 4 where $\beta_{\min}^* = \beta_{\min,2}^* = 1/2$ and $\beta_{\min,1}^* = 1.3\beta_{\min}^*$ to guarantee asymptotic recovery is possible in all four scenarios. Recall that $\gamma^* \in (1/2, 1)$ and $\delta > 0$ is arbitrarily close to 1. We set $\gamma^* = 3/4$ and $\delta = 0.99$ for illustration purposes. Changes in γ^* and δ impact how much the ratios differ between scenarios but not their asymptotic behavior. In the informative Scenarios 1 to 3 convergence with $\hat{S}^{ei}$ is much faster than with $\hat{S}^{sd}$ and the ratios vanish. The vanishing is the fastest in Scenario 3, followed by 2 and 1. In Scenario 4, the ratio converges to a constant just below 0.75. In that scenario, where the partition is non-informative of the distribution of inactive

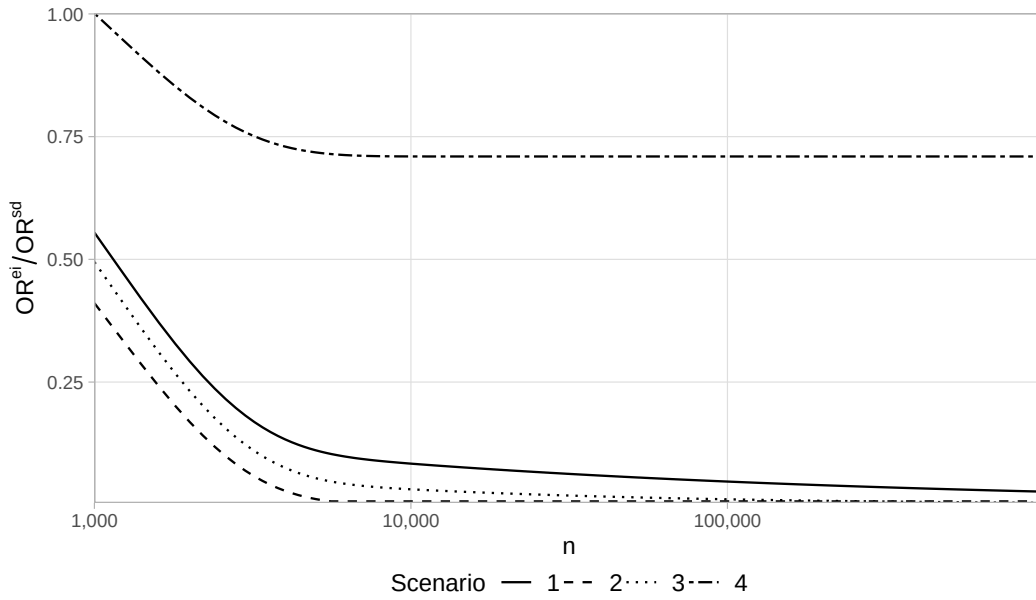


Figure S2: Ratio OR^{ei}/OR^{sd} in Scenario 1 (solid), 2 (dashed), 3 (dotted), and 4 (two-dashed) as in Table S1 (Section S1)

parameters, with $\hat{S}^{ei}$, the oracle is still capable to leverage the difference in signal strength between blocks.

S2 Complementary material on numerical illustrations

Figure S3 shows the average FDR, power and mean squared error in Scenario 1 and 3 of the simulations comparing the adaptive informed selector $\hat{S}^{A,ei}$ to other selection methods in Section 6.1. Recall, Scenario 1 is in informative block partition setting and Scenario 3 a non-informative block partition setting. See Table 1 for a complete description of the scenarios. The results in Scenario 1 confirm the findings in Scenario 2 discussed in Section 6.1. That is, when the block partition is informative and compared to the uninformed $\hat{S}^{A,sd}$, $\hat{S}^{A,ei}$ has equal or lower FDR and higher power. LASSO and EBIC have higher power than $\hat{S}^{A,ei}$ but higher FDR. In terms of mean squared error, $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$ and EBIC perform similarly, and better than SCAD and LASSO. The results in Scenario 3 show that when the block partition is non-informative, $\hat{S}^{A,ei}$ and $\hat{S}^{A,sd}$ perform very similarly.

Figure S4 shows the average runtime of Algorithm 1 to obtain $\hat{S}^{A,ei}$ in Scenario 1–3 as a function of (n, p) (Windows laptop, Intel Core Ultra i7, 32Gb RAM). Runtime increased with the dimension but remained below 20 seconds in all the asymptotic regimes considered.

Figure S5 shows the average FDR, power and mean squared error in Scenario 1 to 3 of the simulations comparing Trans-s-ell0 to other selection methods in Section 6.2. Recall, Scenario 1 and 2 are examples of relevant transfer with respectively complete and moderate overlap of source and target supports. Scenario 3 is an example of irrelevant transfer with no overlap between source and target supports. Again, when the transfer is relevant and compared to the uninformed $\hat{S}^{A,sd}$, Trans-s-ell0 has equally low or lower FDR and higher power. Compared to EBIC, Trans-s-ell0 has higher power in Scenario 1 and similar power

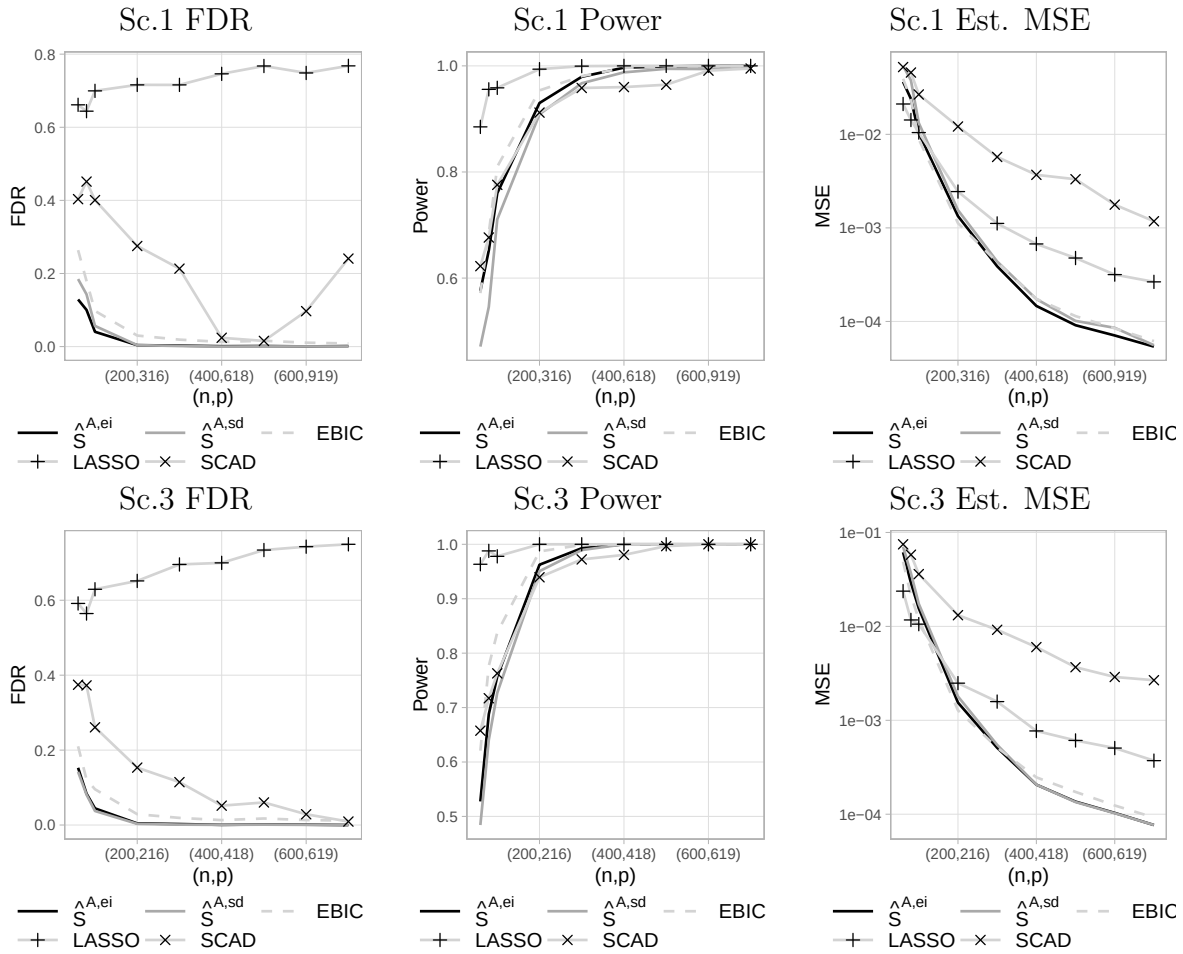


Figure S3: FDR (left), power (middle) and estimation MSE (right) with $\hat{S}^{A,ei}$, $\hat{S}^{A,sd}$, EBIC, LASSO and SCAD in Scenario 1 (top) and 3 (bottom) as in Table 1 (Section 6.1)

in Scenario 2 and has lower FDR in both scenarios. Compared to LASSO, Trans-s-ell0 has lower power but lower FDR in both Scenario 1 and 2. In terms of MSE, Trans-s-ell0 outperforms all the other selection methods in both Scenario 1 and 2. In Scenario 3, Trans-s-ell0 performs very similarly to $\hat{S}^{A,sd}$ except it has slightly more FDR for small sample sizes.

S3 Proofs

Section S3.1 presents results auxiliary to the rest of the proofs, as well as the proofs of that auxiliary results. Section S3.2 the proofs of Section 3 and Section S3.3 the proofs of Sections 4 and 5.

S3.1 Auxiliary results

In this section we collect some technical results auxiliary to the rest of the proofs.

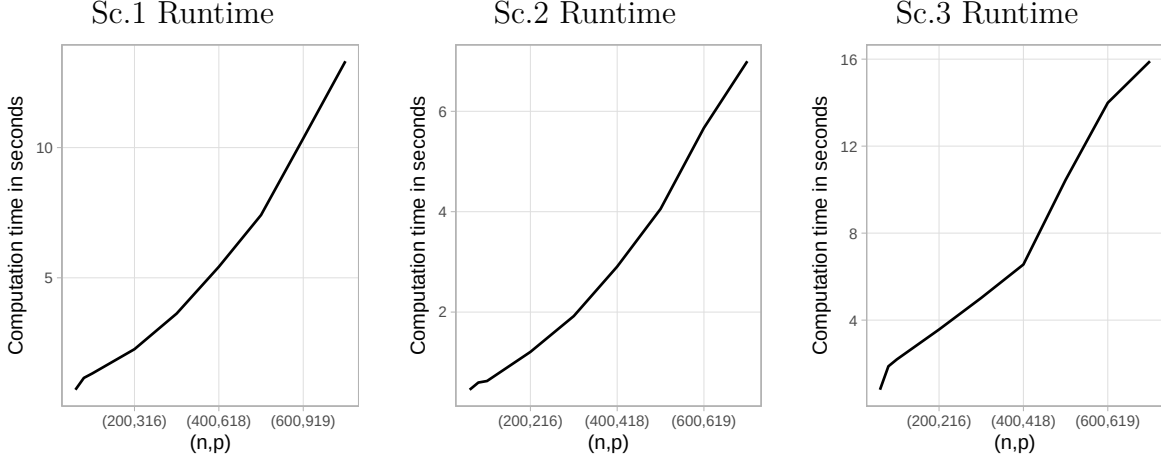


Figure S4: Average runtime of Algorithm 1 to obtain $\hat{S}^{A,ei}$ in Scenario 1 (left), 2 (middle) and 3 (right) as in Table 1 (Section 6.1)

S3.1.1 Tail bounds

Lemma S3.1. *If $y \sim N(\mu, \sigma^2)$ and $a \leq |\mu|$,*

$$P(|y| > a) = P\left(z < \frac{|\mu| - a}{\sigma}\right) + P\left(z < -\frac{a + |\mu|}{\sigma}\right)$$

Proof. We have

$$P(|y| > a) = P(y > a) + P(y < -a) = P\left(z > \frac{a - \mu}{\sigma}\right) + P\left(z < -\frac{a + \mu}{\sigma}\right)$$

where $z \sim N(0, 1)$. If $\mu > 0$, $P(z > \frac{a - \mu}{\sigma}) = P(z > \frac{a - |\mu|}{\sigma}) = P(z < \frac{|\mu| - a}{\sigma})$ by symmetry of the standard Gaussian, and $P(z < -\frac{a + \mu}{\sigma}) = P(z < -\frac{a + |\mu|}{\sigma})$, then $P(|y_i| > a) = P(z < \frac{|\mu| - a}{\sigma}) + P(z < -\frac{a + |\mu|}{\sigma})$. If $\mu \leq 0$, then $P(z > \frac{a - \mu}{\sigma}) = P(z > \frac{a + |\mu|}{\sigma}) = P(z < -\frac{a + |\mu|}{\sigma})$, by symmetry of the standard Gaussian, and $P(z < -\frac{a + \mu}{\sigma}) = P(z < \frac{|\mu| - a}{\sigma})$. Then for any μ , $P(|y_i| > a) = P(z < \frac{|\mu| - a}{\sigma}) + P(z < -\frac{a + |\mu|}{\sigma})$. $\square$

Lemma S3.2. *Let $W \sim \chi_\nu^2(\mu)$ with $\mu \geq 0$, then for any $w > \mu + \nu$*

$$P(W > w) \leq e^{-\left(\frac{w + \mu}{2} - \sqrt{2w(2\mu + \nu) - 2\mu\nu - \nu^2}\right)}.$$

Moreover, assume w , ν and μ are functions of n such that w is increasing, $\nu = o(w)$, and $\mu = o(w)$. Then, for any $\phi \in (0, 1)$ and n large enough

$$P(W > w) \leq e^{-\phi \frac{w}{2}}.$$

Proof. By Birgé (2001), Lemma 8.1 we have that for any $x > 0$

$$P(W > (\nu + \mu) + 2\sqrt{(\nu + 2\mu)x} + 2x) \leq e^{-x}.$$

The function $f : x \mapsto (\nu + \mu) + 2\sqrt{(\nu + 2\mu)x} + 2x$ is one-to-one between $\mathbb{R}^+$ and $(\nu + \mu, \infty)$. Hence, we have that for any $w > \mu + \nu$,

$$P(W > w) \leq e^{-f^{-1}(w)} = e^{-\left(\frac{w+\mu}{2} - \frac{1}{2}\sqrt{2w(2\mu+\nu)-2\mu\nu+\nu^2}\right)}.$$

Observe that

$$\frac{w+\mu}{2} - \frac{1}{2}\sqrt{2w(2\mu+\nu)-2\mu\nu-\nu^2} = \frac{w}{2} \left(1 + \frac{\mu}{w} - \frac{1}{2}\sqrt{\frac{8(2\mu+\nu)}{w} \left(1 - \frac{2\nu\mu - \nu^2}{2(2w\mu + w\nu)}\right)}\right).$$

Since $\nu = o(w)$ and $\mu = o(w)$ by assumption, we have $\frac{w+\mu}{2} - 2^{-1}\sqrt{2w(2\mu+\nu)-2\mu\nu-\nu^2} = \frac{w}{2}(1 + o(1))$. Therefore, for any $\phi \in (0, 1)$ and every n large enough.

$$P(W > w) \leq e^{-\phi \frac{w}{2}}.$$

□

Lemma S3.3. *Let $W \sim \chi_\nu^2(\mu)$ with $\mu > 0$. For any $w < \mu$,*

$$P(W < w) \leq \frac{e^{-\frac{1}{2}(\sqrt{\mu}-\sqrt{w})^2}}{(\mu/w)^{\nu/4}}.$$

Proof. The result follows directly from [Russell \(2022\)](#), Lemma S2. □

Lemma S3.4. *Let $W \sim \chi_\eta^2(\mu)$ with $\mu \geq 0$. Assume that h, c, η and μ are functions of n such that h is positive and increasing, c is positive with $\lim_{n \rightarrow \infty} c = l \in (0, \infty)$, $\eta = o(c \ln(h))$, and $\mu = o(c \ln(h))$. Let $\bar{u}, \underline{u}$ in $(0, 1)$ such that $1 > \bar{u} > \underline{u} \geq \left(1 + h^{\frac{\psi c}{2}} e^{-(\eta+\mu)}\right)^{-1}$ where $\psi \in (0, \min\{1, 2/l\})$, then for every n large enough, we have*

$$\int_{\underline{u}}^{\bar{u}} P\left(W > c \ln\left(\frac{h}{1/u-1}\right)\right) du \leq \frac{1}{h^{\frac{\psi c}{2}}} \left(\bar{u} - \underline{u} + \ln\left(\frac{\bar{u}}{\underline{u}}\right)\right).$$

Proof. For any $u \in [\underline{u}, \bar{u}]$, we have $c \ln\left(\frac{h}{1/u-1}\right) \geq c \ln\left(\frac{h}{1/\underline{u}-1}\right)$. Since $\underline{u} \geq \left(1 + h^{\frac{\psi c}{2}} e^{-(\eta+\mu)}\right)^{-1}$ by assumption we also have that, for any $u \in (\underline{u}, \bar{u})$,

$$c \ln\left(\frac{h}{1/u-1}\right) \geq c\left(1 - \frac{\psi c}{2}\right) \ln(h) + c(\eta + \mu).$$

It follows,

$$\frac{\eta}{c \ln\left(\frac{h}{1/u-1}\right)} \leq \frac{\eta}{c\left(1 - \frac{\psi c}{2}\right) \ln(h) + c(\eta + \mu)} \quad \text{and} \quad \frac{\mu}{c \ln\left(\frac{h}{1/u-1}\right)} \leq \frac{\mu}{c\left(1 - \frac{\psi c}{2}\right) \ln(h) + c(\eta + \mu)}.$$

By assumption $\eta = o(c \ln(h))$ and $\mu = o(c \ln(h))$, then for any $u \in (\underline{u}, \bar{u})$, $\eta = o\left(c \ln\left(\frac{h}{1/u-1}\right)\right)$ and $\mu = o\left(c \ln\left(\frac{h}{1/u-1}\right)\right)$. By Lemma S3.2 with $\phi = \psi \in (0, 1)$, for every n large enough,

$$\int_{\underline{u}}^{\bar{u}} P\left(W > c \ln\left(\frac{h}{1/u-1}\right)\right) du < \frac{1}{h^{\frac{\psi c}{2}}} \int_{\underline{u}}^{\bar{u}} (1/u - 1)^{\frac{\psi c}{2}} du. \quad (\text{S1})$$

Applying the change of variables $v = 1/u - 1$ to the integral on the right-hand side above gives

$$\int_{\underline{u}}^{\bar{u}} (1/u - 1)^{\frac{\psi c}{2}} du = \int_{1/\bar{u}-1}^{1/\underline{u}-1} \frac{v^{\frac{\psi c}{2}}}{(v+1)^2} dv. \quad (\text{S2})$$

Rewrite $v^{\frac{\psi c}{2}} = (v - 1 + 1)^{\frac{\psi c}{2}}$ where, since $\psi < 2/l$ by assumption, $\psi c/2 \leq 1$ for n large enough. Since $\bar{u} < 1$, we have that for any $v > 1/\bar{u} - 1$, $v - 1 > -1$. Note that for any $x \geq -1$ and $r \in [0, 1]$ $(1+x)^r \leq 1+rx$. Then, for any $v > 1/\bar{u} - 1$, $v^{\frac{\psi c}{2}} = (v - 1 + 1)^{\frac{\psi c}{2}} \leq 1 + \frac{\psi c}{2}(v - 1) \leq 1 + \frac{\psi c}{2}(v + 1)$. Applying this last inequality to the right-hand side in (S2) gives

$$\int_{\underline{u}}^{\bar{u}} (1/u - 1)^{\frac{\psi c}{2}} du < \int_{1/\bar{u}-1}^{1/\underline{u}-1} \frac{1}{(v+1)^2} + \frac{\psi c}{2(v+1)} dv = \bar{u} - \underline{u} + \frac{\psi c}{2} \ln\left(\frac{\bar{u}}{\underline{u}}\right). \quad (\text{S3})$$

The result follows inputting the bound from (S3) in (S1) and using that $\psi c/2 \leq 1$ for every n large enough and $\ln(\bar{u}/\underline{u}) \geq 0$ ($\underline{u} \leq \bar{u}$). $\square$

Lemma S3.5. *Let $W \sim \chi_{\eta}^2(\mu)$ with $\mu \geq 0$. Assume that h, c, η and μ are functions of n such that h is positive and increasing, c is positive such that $\lim_{n \rightarrow \infty} c = l \in (0, \infty)$, $\eta = o(\ln(h))$, and $\mu = o(\ln(h))$. Then for any $\alpha \in (0, \min\{1, l/2\})$ and every large enough n , we have*

$$\int_0^1 P\left(W > c \ln\left(\frac{h}{1/u - 1}\right)\right) du = o(h^{-\alpha}).$$

Proof. Since a probability is bounded by 1, for any $a \in (0, 1)$,

$$\int_0^1 P\left(W > c \ln\left(\frac{h}{1/u - 1}\right)\right) du \leq 2a + \int_a^{1-a} P\left(W > c \ln\left(\frac{h}{1/u - 1}\right)\right) du. \quad (\text{S4})$$

Take $a = \left(1 + h^{\frac{\psi c}{2}} e^{-(\eta+\mu)}\right)^{-1}$ where $\psi \in (2\alpha/l, \min\{1, 2/l\})$. Since, by assumption, $\eta = o(c \ln(h))$ and $\mu = o(c \ln(h))$, by Lemma S3.4 with $\underline{u} = a$ and $\bar{u} = 1 - a$, we have

$$\int_0^1 P\left(W > c \ln\left(\frac{h}{1/u - 1}\right)\right) du \leq \frac{2}{1 + h^{\frac{\psi c}{2}} e^{-(\eta+\mu)}} + \frac{1 - 2a + \ln(h^{\frac{\psi c}{2}} e^{-(\eta+\mu)})}{h^{\frac{\psi c}{2}}}.$$

Since $1 + h^{\frac{\psi c}{2}} e^{-(\eta+\mu)} > h^{\frac{\psi c}{2}} e^{-(\eta+\mu)}$ and $1 - 2a - \eta + \mu \leq \ln\left(h^{\frac{\psi c}{2}}\right)$ for every n large enough, we have

$$\int_0^1 P\left(W > c \ln\left(\frac{h}{1/u - 1}\right)\right) du \leq h^{-\frac{\psi c}{2}} (2e^{\eta+\mu} + 2 \ln(h^{\frac{\psi c}{2}}))$$

We have that

$$\frac{h^{-\frac{\psi c}{2}} 2e^{\eta+\mu}}{h^{-\alpha}} = e^{-\left(\frac{\psi c}{2} - \alpha\right) \ln(h) + \ln(2) + \eta + \mu} = e^{-\left(\frac{\psi c}{2} - \alpha\right) \ln(h) \left(1 - \frac{\ln(2)}{(\psi c/2 - \alpha) \ln(h)} - \frac{\eta + \mu}{(\psi c/2 - \alpha) \ln(h)}\right)} \quad (\text{S5})$$

and similarly that

$$\frac{h^{-\frac{\psi c}{2}} 2 \ln(h^{\frac{\psi c}{2}})}{h^{-\alpha}} = e^{-\left(\frac{\psi c}{2} - \alpha\right) \ln(h) \left(1 - \frac{\ln(\psi c \ln(h)/2)}{(\psi c/2 - \alpha) \ln(h)} - \frac{\ln(2)}{(\psi c/2 - \alpha) \ln(h)}\right)}. \quad (\text{S6})$$

Since $\psi > 2\alpha/l$, for n large enough $\alpha < \psi c/2$, and by assumption h is increasing, $\eta = o(\ln(h))$ and $\mu = o(\ln(h))$, both expressions in (S5) and (S6) then vanish as n grows. Hence,

$$\int_0^1 P\left(W > c \ln\left(\frac{h}{1/u - 1}\right)\right) du = o(h^{-\alpha}).$$

□

S3.1.2 Bounds related to model normalized scores

Let $NC(M)$ be the normalized score for model M defined in (9), $\mathcal{M}$ the set of models under consideration. Lemma S3.6 generalizes Lemma 3.2 and shows that the probability of not selecting a set of models is bounded above by the expected sum of the normalized scores of the models outside the set. Lemma S3.7 shows the difference in squared errors for two nested models is chi-squared distributed. Lemma S3.8 and Lemma S3.9 give bounds on the noncentrality parameter of that chi-squared distribution. Lemma S3.10 provides a bound on the expected normalized score of any $M \in \mathcal{M}$ based on the pairwise comparison $C(T) - C(M)$ (cf (9)) for some $T \subseteq S$. It essentially shows that, if the block penalties diverge and the signals in $M \setminus T$ are small, $NC(M)$ is small. Prior to stating Lemma S3.10 we outline the proof strategy.

Lemma S3.6. *For $\hat{S}^{ei}$ as in (8) and any k models $M^1, \dots, M^k$*

$$P(\hat{S}^{ei} \notin \{M^1, \dots, M^k\}) \leq (k+1) \sum_{M \in \mathcal{M} \setminus \{M^1, \dots, M^k\}} \mathbb{E}(NC(M)).$$

Proof. Suppose that $NC(M^1) + \dots + NC(M^k) > \frac{k}{k+1}$ then for any $M \notin \{M^1, \dots, M^k\}$, we have that

$$NC(M) = 1 - \sum_{M' \neq M} NC(M') < 1 - \sum_{M' \in \{M^1, \dots, M^k\}} NC(M') < \frac{1}{k+1}.$$

In addition, if $NC(M^1) + \dots + NC(M^k) > \frac{k}{k+1}$ then necessarily $\max_{l=1, \dots, k} NC(M_l) > \frac{1}{k+1} > NC(M)$ for any $M \notin \{M^1, \dots, M^k\}$, and therefore $\hat{S}^{ei} \in \{M^1, \dots, M^k\}$. Consequently,

$$P(\hat{S}^{ei} \notin \{M^1, \dots, M^k\}) \leq P\left(NC(M^1) + \dots + NC(M^k) \leq \frac{k}{k+1}\right).$$

Moreover, we have

$$P\left(NC(M^1) + \dots + NC(M^k) \leq \frac{k}{k+1}\right) = P\left(\sum_{M \in \mathcal{M} \setminus \{M^1, \dots, M^k\}} NC(M) \geq \frac{1}{k+1}\right).$$

The result follows from the Markov's inequality applied to the right-hand side above. □

Lemma S3.7. *Let M, Q be any two nested models such that $M \subseteq Q$. Then*

$$L_{QM} = \|\mathbf{X}_Q \tilde{\boldsymbol{\beta}}^{(Q)}\|^2 - \|\mathbf{X}_M \tilde{\boldsymbol{\beta}}^{(M)}\|^2 \sim \chi_{|Q \setminus M|}^2(\mu_{QM}),$$

where $\chi_k^2(\mu)$ denotes, when $\mu > 0$, the noncentral chi-squared distribution with k degrees of freedom and noncentrality parameter μ and, when $\mu = 0$, the chi-squared distribution with k degrees of freedom χ_k^2 . The parameter μ_{QM} is given by

$$\mu_{QM} := \|(I_n - P_M)\mathbf{X}_{Q \setminus M}\boldsymbol{\beta}_{Q \setminus M}^*\|^2 \quad (\text{S7})$$

where $P_M = \mathbf{X}_M (\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{X}_M^\top$.

Proof. This result follows directly from Lemma S7 in Rossell (2022) taking $\phi^* = 1$. $\square$

Lemma S3.8. For any $T \subseteq S$ and $M \in \mathcal{M}$ such that $T \not\subseteq M$, let $Q_T = M \cup T$. The non-centrality parameter defined in (S7) satisfies:

$$\mu_{Q_TM} \geq n \rho(\mathbf{X}) \sum_{j=1}^b |T_j \setminus M_j| \min_{i \in T_j \setminus M_j} \beta_i^{*2}. \quad (\text{S8})$$

Proof. The non-centrality parameter μ_{Q_TM} , as defined in (S7), satisfies

$$\begin{aligned} \mu_{Q_TM} &= \|(I_n - P_M)\mathbf{X}_{Q_T \setminus M}\boldsymbol{\beta}_{Q_T \setminus M}^*\|^2 \\ &= \|(I_n - P_M)\mathbf{X}_{T \setminus M}\boldsymbol{\beta}_{T \setminus M}^*\|^2 \\ &= n \boldsymbol{\beta}_{T \setminus M}^{*\top} \left(\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M} \right) \boldsymbol{\beta}_{T \setminus M}^* \\ &\geq n \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M} \right) \|\boldsymbol{\beta}_{T \setminus M}^*\|^2, \end{aligned}$$

where the second equality follows from observing that $Q_T \setminus M = T \setminus M$. Since T is a subset of S , by reordering columns, $\mathbf{X}_{S \setminus M} = [\mathbf{X}_{T \setminus M}, \mathbf{X}_{S \setminus (M \cup T)}]$ and therefore $\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M}$ is a principal submatrix of $\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M}$. Hence, Cauchy's interlacing theorem gives that $\lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{T \setminus M}^\top (I_n - P_M) \mathbf{X}_{T \setminus M} \right) \geq \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M} \right)$. Finally, by definition of $\rho(\mathbf{X})$ in Assumption (A2) we have that $\lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{S \setminus M}^\top (I_n - P_M) \mathbf{X}_{S \setminus M} \right) \geq \rho(\mathbf{X})$, and further noting that $\|\boldsymbol{\beta}_{T \setminus M}^*\|^2 \geq \sum_{j=1}^b |T_j \setminus M_j| \min_{i \in T_j \setminus M_j} \beta_i^{*2}$ gives the desired result. $\square$

Lemma S3.9. For any $T \subseteq S$ and $M \in \mathcal{M}$, let $Q_T = M \cup T$, and $\mu_{Q_T T}$ as defined in (S7), then,

$$\mu_{Q_T T} \leq n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \max_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2} \quad \text{where} \quad \bar{\lambda} := \lambda_{\max} \left(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S \right) \quad (\text{S9})$$

and

$$\mu_{Q_T T} \geq n \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S \right) \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \min_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2} \quad (\text{S10})$$

Proof. Using the definition of $\mu_{Q_T T}$ in (S7), we have that

$$\begin{aligned} \mu_{Q_T T} &= \boldsymbol{\beta}_{Q_T \setminus T}^{*\top} \mathbf{X}_{Q_T \setminus T}^\top (I_n - P_T) \mathbf{X}_{Q_T \setminus T} \boldsymbol{\beta}_{Q_T \setminus T}^* \\ &= \boldsymbol{\beta}_{M \setminus T}^{*\top} \mathbf{X}_{M \setminus T}^\top (I_n - P_T) \mathbf{X}_{M \setminus T} \boldsymbol{\beta}_{M \setminus T}^* \\ &= \boldsymbol{\beta}_{(S \cap M) \setminus T}^{*\top} \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T} \boldsymbol{\beta}_{(S \cap M) \setminus T}^* \end{aligned} \quad (\text{S11})$$

where the second equality follows from $Q_T \setminus T = M \setminus T$ and the third equality from $\beta_{M \setminus S}^* = 0$. We start by showing the upper bound in (S9). Denote for any square matrix A , its largest eigenvalue $\lambda_{\max}(A)$. By (S11), we have that

$$\mu_{Q_T T} \leq n \lambda_{\max} \left(\frac{1}{n} \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T} \right) \|\beta_{(S \cap M) \setminus T}^*\|^2.$$

Let $B := \frac{1}{n} \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T}$, $C := \frac{1}{n} \mathbf{X}_{(S \cap M) \cup T}^\top \mathbf{X}_{(S \cap M) \cup T}$ and $D := \frac{1}{n} \mathbf{X}_T^\top \mathbf{X}_T$. D is a principal submatrix of C and B is the Schur complement of D of C . The inverse B^{-1} is then a principal submatrix of C^{-1} , and by Cauchy's interlacing theorem we have that $\lambda_{\min}(B^{-1}) \geq \lambda_{\min}(C^{-1})$ and then $\lambda_{\max}(B) \leq \lambda_{\max}(C)$. Since $T \subseteq S$ by assumption, we also have that $(S \cap M) \cup T \subseteq S$, then by interlacing again $\lambda_{\max}(C) \leq \lambda_{\max} \left(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S \right) = \bar{\lambda}$. The upper bound in (S9) follows from the latter inequality and also observing that $\|\beta_{(S \cap M) \setminus T}^*\|_2^2 \leq \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \max_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2}$.

We now derive the lower bound in (S10). By (S11), we have that

$$\mu_{Q_T T} \geq n \lambda_{\min} \left(\frac{1}{n} \mathbf{X}_{(S \cap M) \setminus T}^\top (I_n - P_T) \mathbf{X}_{(S \cap M) \setminus T} \right) \|\beta_{(S \cap M) \setminus T}^*\|_2^2.$$

Recall that B^{-1} is a principal submatrix of C^{-1} , hence by interlacing $\lambda_{\max}(B^{-1}) \leq \lambda_{\max}(C^{-1})$ and $\lambda_{\min}(B) \geq \lambda_{\min}(C)$. Since $T \subseteq S$ by assumption, we have that $(S \cap M) \cup T \subseteq S$, and hence $\lambda_{\min}(C) \geq \lambda_{\min}(\frac{1}{n} \mathbf{X}_S^\top \mathbf{X}_S)$. The bound in (S10) is obtained by using the latter inequality and noting that also $\|\beta_{(S \cap M) \setminus T}^*\|_2^2 \geq \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j| \min_{i \in (S_j \cap M_j) \setminus T_j} \beta_i^{*2}$. $\square$

Lemma S3.10 next gives a bound on the expected normalized score of any $M \in \mathcal{M}$. We outline now the strategy to prove it. Recall that by Lemma 3.2(ii), for any M, M' , the normalized score $NC(M)$ is bounded by a simple function of $C(M') - C(M)$. In the next lemma we take $M' = T$ where $T \subseteq S$. By the definition in (9), $C(T) - C(M) = \frac{1}{2} L_{TM} + \Delta_{MT}$, where for any two models $M, T \subseteq V$, we denote

$$\Delta_{MT} := \sum_{j=1}^b \kappa_j (|M_j| - |T_j|) \quad \text{and} \quad L_{TM} := \|\mathbf{X}_T \tilde{\beta}^{(T)}\|^2 - \|\mathbf{X}_M \tilde{\beta}^{(M)}\|^2. \quad (\text{S12})$$

To lower bound $C(T) - C(M)$, we use that L_{TM} can be expressed in terms of chi-squared variables. The idea is to take the union model $Q_T = T \cup M$, and to note that $L_{TM} = L_{Q_T M} - L_{Q_T T}$. Lemma S3.7 is used to bound $L_{Q_T M}$, $L_{Q_T T}$, and hence also L_{TM} .

If $M \supset T$, then $Q_T = M$ and $-L_{TM} = L_{Q_T T}$ and by Lemma S3.7, $-L_{TM} \sim \chi_{|Q_T \setminus T|}^2(\mu_{Q_T T})$, which has expectation $|Q_T \setminus T| + \mu_{Q_T T}$. It also holds that $\Delta_{MT} > 0$. Using properties of chi-squared distribution, we show that if one sets large enough κ_j (and thus Δ_{MT}), then $C(T) - C(M)$ is also large and $NC(M)$ vanishes. If $M \subset T$, then $Q_T = T$, $L_{TM} = L_{Q_T M}$, and by Lemma S3.7 $L_{TM} \sim \chi_{|Q_T \setminus M|}^2(\mu_{Q_T M})$, which has expectation $|Q_T \setminus M| + \mu_{Q_T M}$. If the noncentrality parameter $\mu_{Q_T M}$ is large enough, then L_{TM} and $C(T) - C(M)$ are also large, and, by the properties of the noncentral chi-squared distribution, $NC(M)$ vanishes. By Lemma S3.8, large $\mu_{Q_T M}$ can be achieved by setting a condition on signal strength and $\rho(\mathbf{X})$.

If $M \neq T$ is such that $M \not\supset T$ and $M \not\subset T$, simultaneously large enough κ_j and $\mu_{Q_T M}$ guarantee that $C(T) - C(M)$ is also large, and that $NC(M)$ vanishes.

Lemma S3.10. For any $T \subseteq S$ and $M \in \mathcal{M} \setminus \{T\}$, denote $Q_T = M \cup T$ and $A_T := \gamma\Delta_{MT} + \frac{1-\gamma}{6}\mu_{Q_TM}$ (cf (S12) and (S7)). Suppose that, for some $\gamma \in (1/2, 1]$, it holds that $A_T > 0$, $|M \setminus T| = o(A_T)$, and $\mu_{Q_T T} = o(A_T)$. For any $\psi \in (0, 1)$ and every n large enough,

$$\mathbb{E}(NC(M)) \leq e^{-\psi A_T}.$$

Proof. Since $M \in \mathcal{M} \setminus \{T\}$, by Lemma 3.2 (ii), $NC(M) \leq (1 + e^{C(T)-C(M)})^{-1} \in [0, 1]$. The first step of the proof is to use that for any random variable $Z \geq 0$ we have $\mathbb{E}(Z) = \int_0^\infty P(Z > u)du$, so that

$$\begin{aligned} \mathbb{E}(NC(M)) &\leq \int_0^1 P\left((1 + e^{C(T)-C(M)})^{-1} \geq u\right) du \\ &= \int_0^1 P\left(C(T) - C(M) \leq \ln\left(\frac{1}{u} - 1\right)\right) du \\ &= \int_0^1 P\left(-\frac{1}{2}L_{TM} \geq \Delta_{MT} - \ln\left(\frac{1}{u} - 1\right)\right) du. \end{aligned}$$

The second step of the proof is to use the union bound to upper bound the probability in the integrand above. Let $Q_T = M \cup T$ and recall that $L_{TM} = L_{Q_TM} - L_{Q_T T}$. For any γ

$$-\frac{1}{2}L_{TM} - (\Delta_{MT} - \ln\left(\frac{1}{u} - 1\right)) = \left(\frac{1}{2}L_{Q_T T} - \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right)\right) - \left(\frac{1}{2}L_{Q_TM} + (1-\gamma)\Delta_{MT}\right).$$

Observe that for any random variables U, V , and any $\epsilon, \gamma' \geq 0$, the event $\{U - V \geq 0\}$ implies $\{U \geq \gamma'\epsilon\} \cup \{V < \gamma'\epsilon\}$. Let $U = \frac{1}{2}L_{Q_T T} - \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right)$ and $V = \frac{1}{2}L_{Q_TM} + (1-\gamma)\Delta_{MT}$. Take $\epsilon = \mu_{Q_TM}$ and $\gamma' = \frac{1}{6}(1-\gamma)$, and observe that $A_T := \gamma\Delta_{MT} + \frac{1-\gamma}{6}\mu_{Q_TM} = \gamma\Delta_{MT} + \gamma'\mu_{Q_TM}$. We then have

$$\begin{aligned} \{U \geq \gamma'\epsilon\} &= \left\{\frac{1}{2}L_{Q_T T} \geq \ln\left(\frac{e^{\gamma\Delta_{MT}}}{\frac{1}{u} - 1}\right) + \gamma'\mu_{Q_TM}\right\} = \left\{\frac{1}{2}L_{Q_T T} \geq \ln\left(\frac{e^{A_T}}{\frac{1}{u} - 1}\right)\right\} \\ \{V < \gamma'\epsilon\} &= \left\{\frac{1}{2}L_{Q_TM} < -(1-\gamma)\Delta_{MT} + \gamma'\mu_{Q_TM}\right\} = \left\{\frac{1}{2}L_{Q_TM} < \gamma'(\mu_{Q_TM} - 6\Delta_{MT})\right\}. \end{aligned}$$

By the union bound we have that

$$\mathbb{E}(NC(M)) \leq \int_0^1 P\left(\frac{1}{2}L_{Q_T T} \geq \ln\left(\frac{e^{A_T}}{\frac{1}{u} - 1}\right)\right) du + P\left(\frac{1}{2}L_{Q_TM} < \gamma'(\mu_{Q_TM} - 6\Delta_{MT})\right). \quad (\text{S13})$$

The third and final step of the proof is to upper bound each of the terms in the right-hand side of (S13). The intuition is that both T and M are nested within Q_T , and therefore $L_{Q_T T}$ and L_{Q_TM} follow chi-squared distributions. We first bound the first term. If $M \subset T$ then $Q_T = T$, $L_{Q_T T} = 0$, and this term is zero. Suppose now that $M \not\subset T$. Then, by Lemma S3.7, $L_{Q_T T} \sim \chi_{|Q_T \setminus T|}^2(\mu_{Q_T T})$ with $|Q_T \setminus T| = |M \setminus T|$. By assumption, $A_T > 0$, $|M \setminus T| = o(\ln(e^{A_T}))$ and $\mu_{Q_T T} = o(\ln(e^{A_T}))$, then by Lemma S3.5, with $c = 2$ and $\alpha \in (\psi, 1)$, and every n large enough,

$$\int_0^1 P\left(L_{Q_T T} > 2 \ln\left(\frac{e^{A_T}}{1/u - 1}\right)\right) du < e^{-\alpha A_T}. \quad (\text{S14})$$

We now bound the second term in (S13). If $M \supset T$, then $Q_T = M$, $L_{Q_TM} = 0$, $\mu_{Q_TM} = 0$, and this term is zero. Alternatively, if $M \not\supset T$ then, by Lemma S3.7, $L_{Q_TM} \sim \chi^2_{|Q_T \setminus M|}(\mu_{Q_TM})$ with $|Q_T \setminus M| = |T \setminus M|$. Clearly, when $\mu_{Q_TM} \leq 6\Delta_{MT}$ this term is also zero, so suppose that $\mu_{Q_TM} > 6\Delta_{MT}$. We have

$$P(L_{Q_TM} < 2\gamma'(\mu_{Q_TM} - 6\Delta_{MT})) \leq P(L_{Q_TM} < 2\gamma'\mu_{Q_TM})$$

and, by Lemma S3.3 and using that $\gamma' \in (0, \frac{1}{12})$, we obtain that

$$P(L_{Q_TM} < 2\gamma'\mu_{Q_TM}) \leq (2\gamma')^{\frac{|T \setminus M|}{4}} e^{-\frac{1}{2}(1-\sqrt{2\gamma'})^2 \mu_{Q_TM}} \leq \left(\frac{1}{6}\right)^{\frac{|T \setminus M|}{4}} e^{-\frac{1}{2}(1-\sqrt{2\gamma'})^2 \mu_{Q_TM}}.$$

Since $\mu_{Q_TM} > 6\Delta_{MT}$, we get

$$A_T = \gamma'\mu_{Q_TM} + \gamma\Delta_{MT} < \frac{1-\gamma}{6}\mu_{Q_TM} + \frac{\gamma}{6}\mu_{Q_TM} = \frac{1}{6}\mu_{Q_TM} \leq \frac{1}{2}(1-\sqrt{2\gamma'})^2 \mu_{Q_TM},$$

where the last inequality follows from the fact that $\gamma' \in (0, \frac{1}{12})$. We then get

$$P(L_{Q_TM} < 2\gamma'(\mu_{Q_TM} - 6\Delta_{MT})) \leq \left(\frac{1}{6}\right)^{\frac{|T \setminus M|}{4}} e^{-A_T} \leq e^{-A_T} \leq e^{-\alpha A_T} \quad (\text{S15})$$

Summing the bounds in (S14) and (S15) gives that for every n large enough $\mathbb{E}(NC(M)) < 2e^{-\alpha A_T}$. Since $\psi < \alpha$, for every n large enough, we have that $\mathbb{E}(NC(M)) < e^{-\psi A_T}$ as claimed. $\square$

S3.2 Proofs of Section 3

S3.2.1 Proof of Lemma 3.1

First note that by discarding constant terms

$$\arg \max_{M \in \mathcal{M}} \left\{ \max_{\beta \in \mathcal{L}_M} \ell(\mathbf{y}; \beta) - \sum_{j=1}^b \kappa_j |M_j| \right\} = \arg \min_{M \in \mathcal{M}} \left\{ \min_{\beta \in \mathcal{L}_M} \left\{ \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 \right\} + \sum_{j=1}^b \kappa_j |M_j| \right\}.$$

We also have that

$$\min_{\beta \in \mathcal{L}_M} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 = \frac{1}{2} \|\mathbf{y} - \mathbf{X}_M \tilde{\beta}^{(M)}\|^2 = \frac{1}{2} \|\mathbf{y}\|^2 - \frac{1}{2} \|\mathbf{X}_M \tilde{\beta}^{(M)}\|^2,$$

where in the last equality we used that $\tilde{\beta}^{(M)} = (\mathbf{X}_M^T \mathbf{X}_M)^{-1} \mathbf{X}_M^T \mathbf{y}$. The maximization in (8) can be then replaced with the maximization of $C(M) = \frac{1}{2} \|\mathbf{X}_M \tilde{\beta}^{(M)}\|^2 - \sum_{j=1}^b \kappa_j |M_j|$ which is equivalent to maximizing $NC(M)$.

S3.2.2 Proof of Lemma 3.2

Part (i) follows directly from Lemma S3.6 by taking $\{M^1, \dots, M^k\} = \{S\}$. Part (ii) follows from

$$NC(M) = \left(1 + \sum_{N \neq M} e^{C(N) - C(M)} \right)^{-1} < \left(1 + e^{C(M') - C(M)} \right)^{-1}.$$

S3.2.3 Proof of Theorem 3.3

The proof strategy is to first use Lemma S3.10 with $T = S$ to show that for every $M \neq S$, $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$ for every large enough n and any $\psi \in (0, 1)$, where $A_S = \gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M}$ (cf (S12) and (S7)), $\gamma \in (1/2, 1)$ is defined in Assumption A2 and $Q_S = M \cup S$. Assumptions A1–A2 and the fact that $M \setminus S \subseteq S^C$ ensure the assumptions of Lemma S3.10 are met for any $\beta^* \in \mathfrak{B}$. The second step is to obtain a lower bound for A_S , which gives a new upper bound for $\mathbb{E}(NC(M))$. The final step is to use these bounds to get an upper-bound on $\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M))$ that holds for any $\beta^* \in \mathfrak{B}$ and asymptotically vanishes under Assumptions A1 and A2. We then use Lemma 3.2 to conclude on the vanishing of $\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \neq S)$.

First, to show that $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$ for any $M \in \mathcal{M} \setminus \{S\}$, we show that A_S satisfies the conditions of Lemma S3.10, taking $T = S$. That is, we wish to show that, $A_S > 0$, $|M \setminus S| = o(A_S)$, and $\mu_{Q_S S} = o(A_S)$. Observe that Δ_{MS} , defined in (S12), can be rewritten as $\Delta_{MS} = \sum_{j=1}^b (|M_j \setminus S_j| - |S_j \setminus M_j|) \kappa_j$. By Lemma S3.8 with $T = S$, for any $\beta^* \in \mathfrak{B}$ and every $n \in \mathbb{N}$, we have

$$\begin{aligned} A_S &= \gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M} \\ &\geq \gamma \sum_{j=1}^b |M_j \setminus S_j| \kappa_j + \sum_{j=1}^b |S_j \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^{*2} - \gamma \kappa_j \right) \end{aligned} \quad (\text{S16})$$

Since $M \neq S$, $|M \setminus S| \neq 0$ or $|S \setminus M| \neq 0$, then by Assumptions A1 and A2, for every n large enough, $A_S > 0$. We immediately have $\mu_{Q_S S} = o(A_S)$ because $\beta_{Q_S \setminus S}^* = \beta_{M \setminus S}^* = 0$ (any parameter outside the true support S is by definition 0) and hence $\mu_{Q_S S} = 0$. If $|M \setminus S| = 0$, $|M \setminus S| = o(A_S)$ also immediately. Consider now the case $|M \setminus S| \neq 0$. By Assumption A2, the last term in (S16) is nonnegative, and hence

$$\frac{|M \setminus S|}{A_S} = \frac{|M \setminus S|}{\gamma \Delta_{MS} + \frac{1-\gamma}{6} \mu_{Q_S M}} \leq \left[\gamma \sum_{j=1}^b \frac{|M_j \setminus S_j|}{|M \setminus S|} \kappa_j \right]^{-1} \leq \left[\gamma \min_{j=1, \dots, b} \kappa_j \right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^b \frac{|M_j \setminus S_j|}{|M \setminus S|} = 1$. By Assumption A1 we have that $\min_j \kappa_j \rightarrow \infty$ as $n \rightarrow \infty$, and hence $|M \setminus S| = o(A_S)$. Thus, by Lemma S3.10, for any $\psi \in (0, 1)$ and $\beta^* \in \mathfrak{B}$, and all n large enough, $\mathbb{E}(NC(M)) \leq e^{-\psi A_S}$.

For the second step of the proof, let A_S^* be the lower bound for A_S given in (S16). That is

$$A_S^* := \gamma \sum_{j=1}^b |M_j \setminus S_j| \kappa_j + \sum_{j=1}^b |S_j \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^{*2} - \gamma \kappa_j \right).$$

By (S16), we have, for any $\beta^* \in \mathfrak{B}$ and all n large enough,

$$\mathbb{E}(NC(M)) \leq e^{-\psi A_S^*}. \quad (\text{S17})$$

Assumption A2 implies there exist $g'_j \rightarrow \infty$ such that

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \beta_{\min, j}^{*2} - \kappa_j = \ln(s_j) + g'_j. \quad (\text{S18})$$

Let $\delta \in (0, 1)$ and denote $\bar{m}_j = \max \left\{ \frac{2\ln(p_j - s_j)}{f_j}, \frac{2\ln(s_j)}{g'_j} \right\}$, where f_j is given in Assumption A1. Take $\psi = \max_{j=1, \dots, b} \frac{\xi + \delta + \bar{m}_j}{1 + \bar{m}_j}$ for some $\xi \in (0, 1 - \delta)$ then $\psi \in (0, 1)$ and we have, for every $j = 1, \dots, b$,

$$\psi > \frac{\delta + \frac{2\ln(p_j - s_j)}{f_j}}{1 + \frac{2\ln(p_j - s_j)}{f_j}} = \frac{\delta f_j/2 + \ln(p_j - s_j)}{f_j/2 + \ln(p_j - s_j)} \quad (\text{S19})$$

$$\psi > \frac{\delta + \frac{2\ln(s_j)}{g'_j}}{1 + \frac{2\ln(s_j)}{g'_j}} = \frac{\delta g'_j/2 + \ln(s_j)}{g'_j/2 + \ln(s_j)} \geq \frac{\delta g'_j/2 + \ln(s_j)}{g'_j + \ln(s_j)}. \quad (\text{S20})$$

In Assumption A2, γ is defined as $\gamma := \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j)$, we then have

$$\gamma \kappa_j \geq \frac{1}{2} \left(1 + \frac{\ln(p_j - s_j)}{\kappa_j} \right) \kappa_j = \ln(p_j - s_j) + \frac{1}{2}(\kappa_j - \ln(p_j - s_j)) = \ln(p_j - s_j) + \frac{1}{2}f_j.$$

Hence, by (S19), we have

$$\psi \gamma \kappa_j \geq \psi \left(\ln(p_j - s_j) + \frac{1}{2}f_j \right) \geq \ln(p_j - s_j) + \delta \frac{1}{2}f_j. \quad (\text{S21})$$

Further,

$$\psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^*{}^2 - \gamma \kappa_j \right) \geq \psi \left(\ln(s_j) + g'_j \right) \geq \ln(s_j) + \delta \frac{1}{2}g'_j \quad (\text{S22})$$

where the first inequality follows from (S18) and the second inequality from (S20).

In (S17), $\psi A_S^* = \sum_{j=1}^b |M_j \setminus S_j| \psi \gamma \kappa_j + \sum_{j=1}^b |S_j \setminus M_j| \psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \beta_{\min, j}^*{}^2 - \gamma \kappa_j \right)$. Then by (S21) and (S22), we get, for any $\beta^* \in \mathfrak{B}$,

$$\mathbb{E}(NC(M)) \leq \exp \left\{ - \sum_{j=1}^b |M_j \setminus S_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right) \right\}. \quad (\text{S23})$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M))$ for convenience. Since (S23) holds for any $\beta^* \in \mathfrak{B}$, we have,

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq \sum_{M \in \mathcal{M} \setminus \{S\}} e^{-\sum_{j=1}^b |M_j \setminus S_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

Observe that if $|M_j \setminus S_j| = 0$ and $|S_j \setminus M_j| = 0$ for all j , then $M = S$ and the summand in the right-hand side above is 1. Then by adding and subtracting 1 we get

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq -1 + \sum_{M \in \mathcal{M}} e^{-\sum_{j=1}^b |M_j \setminus S_j| \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b |S_j \setminus M_j| \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

We can split the sum in the right-hand side above into sums over the models that have the same number of inactive variables and missing the same number of truly active variables in

every block. That is, the models M such that for all j , $|M_j \setminus S_j| = u_j$ and $|S_j \setminus M_j| = w_j$ with $u_j \in \{0, \dots, p_j - s_j\}$ and $w_j \in \{0, \dots, s_j\}$. Denote those sums

$$S_{\mathbf{w}}^{\mathbf{u}} = \sum_{M \in \mathcal{M}: \forall j |M_j \setminus S_j| = u_j, |S_j \setminus M_j| = w_j} e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}.$$

We get

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1-s_1} \cdots \sum_{u_b=0}^{p_b-s_b} S_{\mathbf{w}}^{\mathbf{u}}. \quad (\text{S24})$$

The number of models having, for all j , u_j inactive parameters and missing w_j out of the s_j active parameters is $\prod_{j=1}^b \binom{p_j - s_j}{u_j} \binom{s_j}{w_j}$. We thus have

$$\begin{aligned} S_{\mathbf{w}}^{\mathbf{u}} &= \left(\prod_{j=1}^b \binom{p_j - s_j}{u_j} \binom{s_j}{w_j} \right) e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \\ &= \prod_{j=1}^b \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}. \end{aligned}$$

Inputting the expression above in (S24) gives

$$\begin{aligned} \sup_{\beta^* \in \mathfrak{B}} \mathcal{S} &\leq -1 + \sum_{w_1=0}^{s_1} \cdots \sum_{w_b=0}^{s_b} \sum_{u_1=0}^{p_1-s_1} \cdots \sum_{u_b=0}^{p_b-s_b} \prod_{j=1}^b \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \\ &\leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j - s_j} \binom{p_j - s_j}{u_j} e^{-u_j \left(\ln(p_j - s_j) + \delta \frac{f_j}{2} \right)} \right) \left(1 + \sum_{w_j=1}^{s_j} \binom{s_j}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \right) \end{aligned}$$

where the second inequality follows from first factorizing over terms in u_j and w_j and then taking the term in 0 out of every sum. A standard bound on binomial coefficient for $1 \leq k \leq n$ is

$$\binom{n}{k} \leq \left(\frac{ne}{k} \right)^k \leq (ne)^k = e^{k(\ln(n)+1)}. \quad (\text{S25})$$

Then

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j - s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)} \right) \left(1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)} \right). \quad (\text{S26})$$

Denote

$$d_j = e^{1 - \delta \frac{f_j}{2}}, \quad h_j = e^{1 - \delta \frac{g'_j}{2}}$$

where both expressions go to zero as n increases since $f_j \rightarrow \infty$ and $g'_j \rightarrow \infty$. For every j , by the properties of geometric sums, we have

$$\begin{aligned} 1 + \sum_{u_j=1}^{p_j - s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)} &= \frac{1 - d_j^{p_j - s_j + 1}}{1 - d_j} \\ 1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)} &= \frac{1 - h_j^{s_j + 1}}{1 - h_j}. \end{aligned}$$

Since both expressions converge to 1 as n grows, and b is fixed we get that

$$\lim_{n \rightarrow \infty} \sup_{\beta^* \in \mathfrak{B}} \mathcal{S} = \lim_{n \rightarrow \infty} \sup_{\beta^* \in \mathfrak{B}} \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) = 0. \quad (\text{S27})$$

By Lemma 3.2, for any $\beta^* \in \mathfrak{B}$, $P(\hat{S}^{ei} \neq S) \leq 2\mathcal{S}$, then $\lim_{n \rightarrow \infty} \inf_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} = S) = 1$.

S3.2.4 Proof of Theorem 3.4

First, we prove the upper bound on $P(\hat{S}^{ei} \neq S)$ in (10). We assume here A1 and A2. Under the same assumptions, in the proof of Theorem 3.3, the following bound was shown in (S26) for any $\beta^* \in \mathfrak{B}$:

$$\sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq -1 + \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)} \right) \left(1 + \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)} \right). \quad (\text{S28})$$

Denote $S(u_j) = \sum_{u_j=1}^{p_j-s_j} e^{-u_j \left(\delta \frac{f_j}{2} - 1 \right)}$, $S(w_j) = \sum_{w_j=1}^{s_j} e^{-w_j \left(\delta \frac{g'_j}{2} - 1 \right)}$, $d_j = e^{1-\delta \frac{f_j}{2}}$, and $h_j = e^{1-\delta \frac{g'_j}{2}}$. For every j , we have, by the properties of geometric sums:

$$S(u_j) = d_j \frac{1 - d_j^{p_j-s_j}}{1 - d_j}$$

$$S(w_j) = h_j \frac{1 - h_j^{s_j}}{1 - h_j}.$$

Developing the product in the right-hand side in (S28) and reordering the resulting terms gives

$$\sup_{\beta^* \in \mathfrak{B}} \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq -1 + 1 + \sum_{j=1}^b [S(u_j) + S(w_j)] + \mathcal{R}$$

where all the terms in $\mathcal{R}$ are product of two or more of the sums $S(u_1), \dots, S(u_b), S(w_1), \dots, S(w_b)$. Given that $\delta > 0$, $f_j \rightarrow \infty$ and $g'_j \rightarrow \infty$ by assumption, and hence $d_j \rightarrow 0$ and $h_j \rightarrow 0$, the $S(u_j)$ and $S(w_j)$ are smaller than 1 for all sufficiently large n for all j . Then each of the $2^{2b} - 2b - 1$ terms in $\mathcal{R}$ is bounded above by $\sum_{j=1}^b [S(u_j) + S(w_j)]$ and we get, for every n large enough,

$$\sup_{\beta^* \in \mathfrak{B}} \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq (2^{2b} - 2b) \sum_{j=1}^b \left[d_j \frac{1 - d_j^{p_j-s_j}}{1 - d_j} + h_j \frac{1 - h_j^{s_j}}{1 - h_j} \right]. \quad (\text{S29})$$

Denote

$$r = \max_{j=1, \dots, b} \left\{ \frac{1 - d_j^{p_j-s_j}}{1 - d_j}, \frac{1 - h_j^{s_j}}{1 - h_j} \right\}. \quad (\text{S30})$$

By Lemma 3.2, (S29) and (S30), we then obtain:

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \neq S) \leq 2 \sup_{\beta^* \in \mathfrak{B}} \sum_{M \in \mathcal{M} \setminus \{S\}} \mathbb{E}(NC(M)) \leq 2(2^{2b} - 2b)r e \sum_{j=1}^b e^{-\delta \frac{f_j}{2}} + e^{-\delta \frac{g'_j}{2}}$$

By the definition of f_j in Assumption A1, that of g'_j in (S18), and the fact that $2e < 6$, we get, for every n large enough, that

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \neq S) \leq (2^{2b} - 2b)6r \sum_{j=1}^b e^{-\frac{\delta}{2}[\kappa_j - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2}\left[\frac{(1-\gamma)n\rho(\mathbf{X})}{6}\beta_{\min,j}^*{}^2 - \kappa_j - \ln(s_j)\right]}. \quad (\text{S31})$$

We have $\frac{(1-\gamma)n\rho(\mathbf{X})}{6}\beta_{\min,j}^*{}^2 - \kappa_j > \left(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j}\right)^2$ and then,

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \neq S) \leq (2^{2b} - 2b)6r \sum_{j=1}^b e^{-\frac{\delta}{2}[\kappa_j - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2}\left[\left(\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j}\right)^2 - \ln(s_j)\right]}.$$

We have $d_j \rightarrow 0$ and $h_j \rightarrow 0$, then for every n large enough $\frac{1-d_j^{p_j-s_j}}{1-d_j} \rightarrow 1$ and $\frac{1-h_j^{s_j}}{1-h_j} \rightarrow 1$ for all j , and $r \rightarrow 1$. It follows that for any $c > (2^{2b} - 2b)6$ for n large enough the bound in (10) holds.

Second, we prove that, if for all $j = 1, \dots, b$, it holds that

$$\lim_{n \rightarrow \infty} \frac{g(\gamma^*)\sqrt{n\rho(\mathbf{X})/6}\beta_{\min,j}^*}{\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)}} \geq 1 \quad (\text{S32})$$

with $g(\gamma^*) = \sqrt{(-1 + 2\gamma^*)(1 - \gamma^*)}/(1 + \sqrt{2 - 2\gamma^*})$ then κ_j^* defined in (11) satisfies Assumptions A1 and A2. Indeed, under (S32), for every j , there exists a sequence x_j such that $\lim_{n \rightarrow \infty} x_j \geq 1$ and $g(\gamma^*)\sqrt{\frac{n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* = x_j(\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)})$. Let $v_j = x_j\sqrt{(1 - \gamma^*)g(\gamma^*)^{-1}}$, then $\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* = v_j(\sqrt{\ln(p_j - s_j)} + \sqrt{\ln(s_j)})$. Denote $a = \sqrt{\ln(p_j - s_j)}$ and $a' = \sqrt{\ln(s_j)}$. We have

$$\sqrt{\kappa_j^*} = \frac{v_j}{2}(a + a') + \frac{(a^2 - a'^2)}{2v_j(a + a')} = \frac{v_j}{2}(a + a') + \frac{a - a'}{2v_j} = \frac{v_j^2 + 1}{2v_j}a + \frac{v_j^2 - 1}{2v_j}a'.$$

Since $\sqrt{(1 - \gamma^*)g(\gamma^*)^{-1}} > 1$ for $\gamma^* \in (1/2, 1)$ and $\lim_{n \rightarrow \infty} x_j \geq 1$, for n large enough, $v_j > 1$. We also have $a' \geq 0$. It follows that $a'(v_j^2 - 1)/2v_j \geq 0$ for n large enough and

$$\sqrt{\kappa_j^*} = \frac{v_j^2 + 1}{2v_j}a + \frac{v_j^2 - 1}{2v_j}a' \geq \left(\frac{v_j^2 + 1}{2v_j} + 1 - 1\right)a = \left(1 + \frac{(v_j - 1)^2}{2v_j}\right)\sqrt{\ln(p_j - s_j)} \quad (\text{S33})$$

which implies Assumption A1 since $v_j > 1$ for n sufficiently large. We also have

$$\frac{\sqrt{(1 - \gamma^*)n\rho(\mathbf{X})}}{6}\beta_{\min,j}^* - \sqrt{\kappa_j^*} = \frac{v_j^2 - 1}{2v_j}a + \frac{v_j^2 + 1}{2v_j}a' \geq \left(1 + \frac{(v_j - 1)^2}{2v_j}\right)\sqrt{\ln(s_j)}. \quad (\text{S34})$$

Let $\gamma = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j^*)$ as defined in Assumption A2. By (S33), $\gamma = \frac{1}{2}(1 + \max_j (1 + (v_j - 1)^2/(2v_j)))^{-2}$. Since $\lim_{n \rightarrow \infty} x_j \geq 1$, for sufficiently large n , $v_j \geq \sqrt{(1 - \gamma^*)g(\gamma^*)^{-1}}$.

Simple algebra shows the latter inequality implies $\gamma = \frac{1}{2}(1 + \max_j \ln(p_j - s_j)/\kappa_j^*) < \gamma^*$ and $1 - \gamma > 1 - \gamma^*$. Then, by (S34),

$$\frac{\sqrt{(1-\gamma)n\rho(\mathbf{X})}}{6}\beta_{\min,j}^* - \sqrt{\kappa_j^*} \geq \left(1 + \frac{(v_j - 1)^2}{2v_j}\right)\sqrt{\ln(s_j)}$$

for every sufficiently large n . Since $\lim_{n \rightarrow \infty} x_j \geq 1$ and $\sqrt{(1-\gamma^*)g(\gamma^*)^{-1}} > 1$ for $\gamma^* \in (1/2, 1)$, $v_j > 1$ for n sufficiently large and Assumption A2 holds.

Finally, we prove (12). Since Assumptions A1 and A2 hold for the κ_j^* , by the first part of the theorem, there exists $c > 0$ such that, for every n large enough,

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \neq S) \leq c \sum_{j=1}^b e^{-\frac{\delta}{2}[\kappa_j^* - \ln(p_j - s_j)]} + e^{-\frac{\delta}{2}\left[\left(\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j^*}\right)^2 - \ln(s_j)\right]}.$$

Note that the κ_j^* satisfy $\kappa_j^* - \ln(p_j - s_j) = \left(\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* - \sqrt{\kappa_j^*}\right)^2 - \ln(s_j)$ for all j . We then have

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \neq S) \leq 2c \sum_{j=1}^b e^{-\frac{\delta}{2}[\kappa_j^* - \ln(p_j - s_j)]}. \quad (\text{S35})$$

Denote

$d = \frac{1}{2}\sqrt{\frac{6}{(1-\gamma^*)n\rho(\mathbf{X})}}\frac{1}{\beta_{\min,j}^*} \left(\ln(p_j - s_j) - \ln(s_j)\right)$ such that $\sqrt{\kappa_j^*} = \frac{1}{2}\sqrt{\frac{(1-\gamma^*)n\rho(\mathbf{X})}{6}}\beta_{\min,j}^* + d$. Then

$$e^{-\frac{\delta}{2}(\kappa_j^* - \ln(p_j - s_j))} = e^{-\frac{\delta}{2}\left[\frac{(1-\gamma^*)n\rho(\mathbf{X})}{24}\beta_{\min,j}^{*2} + d^2 - \frac{1}{2}(\ln(p_j - s_j) + \ln(s_j))\right]}.$$

By considering separately the two possible maxima in $\ln \max\{p_j - s_j, s_j\}$, we get that

$$\frac{e^{-\frac{\delta}{2}(\kappa_j^* - \ln(p_j - s_j))}}{e^{-\frac{\delta}{2}\left[\frac{(1-\gamma^*)n\rho(\mathbf{X})}{24}\beta_{\min,j}^{*2} - \ln \max\{p_j - s_j, s_j\}\right]}} = e^{-\frac{\delta}{2}\left[d^2 + \frac{1}{2}|\ln(p_j - s_j) - \ln(s_j)|\right]} < 1. \quad (\text{S36})$$

It follows that, by (S35) and (S36),

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \neq S) \leq 2c \sum_{j=1}^b e^{-\frac{\delta}{2}\left[\frac{(1-\gamma^*)n\rho(\mathbf{X})}{24}\beta_{\min,j}^{*2} - \ln \max\{p_j - s_j, s_j\}\right]}$$

which proves (12).

S3.2.5 Proof of Proposition 3.5

Consistent recovery of S implies that for any subset of models $\mathcal{N}$, $\max_{M \in \mathcal{N}} NC(M)/NC(S) < 1$ with probability going to 1 as n grows, where NC is the normalized criterion defined in (9). In Part (i), for every $j = 1, \dots, b$, we obtain a necessary assumption on the κ_j for selection consistency by considering $\mathcal{N} = \mathcal{O}_j$ as defined in (13). In Part (ii), we obtain a necessary assumption on the $\beta_{\min,j}^*$ by considering a model that under-fits by only one variable i with $\beta_i^* = \beta_{\min,j}^*$. In Part (iii), we use (i) and (ii) to get, for every $j = 1, \dots, b$, a necessary assumption on the scaling of $(n, p_j, s_j, \beta_{\min,j}^*)$.

Part (i). The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1$. We have, for any $\beta^* \in \mathfrak{B}$,

$$P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} \frac{NC(M)}{NC(S)} < 1\right) = P\left(\max_{M \in \mathcal{O}_j} e^{C(M)-C(S)} < 1\right).$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (S12), we obtain

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} e^{-(\frac{1}{2}L_{SM} + \Delta_{MS})} < 1\right) = P\left(\max_{M \in \mathcal{O}_j} L_{MS} < 2\kappa_j\right). \quad (\text{S37})$$

Taking square root on both sides of the inequality in the right-most probability in (S37), we have

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} = S) \leq P\left(\max_{M \in \mathcal{O}_j} \sqrt{L_{MS}} < \sqrt{2\kappa_j}\right). \quad (\text{S38})$$

By Lemma S3.7, for every $M \in \mathcal{O}_j$, $L_{MS} \in \chi_1^2$ and there exists $Z_M \sim N(0, 1)$ such that $\sqrt{L_{MS}} = |Z_M|$. Since for every $M \in \mathcal{O}_j$, $|Z_M| \geq Z_M$, we have

$$\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} = S) \leq 1 - P\left(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\kappa_j}\right). \quad (\text{S39})$$

The set $\mathcal{O}_j$ has cardinality $p_j - s_j$ which, by assumption, diverges, then by Theorem 3.4 in Hartigan (2014), for any $\varepsilon > 0$,

$$P\left(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\lambda_j \ln(p_j - s_j)(1 - \varepsilon)}\right) \rightarrow 1,$$

where λ_j is as defined in (14). If $\lim_{n \rightarrow \infty} \frac{\kappa_j}{\lambda_j \ln(p_j - s_j)} < 1$, then $\lim_{n \rightarrow \infty} P\left(\max_{M \in \mathcal{O}_j} Z_M \geq \sqrt{2\kappa_j}\right) = 1$. Hence, by (S39) we have that $\lim_{n \rightarrow \infty} \sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} = S) = 0$, as claimed.

Part (ii). Consider a $\beta^* \in \mathfrak{B}$ such that for some $i_0 \in B_j$, $\beta_{i_0}^* = \beta_{\min,j}^*$ and the model M , that is under-fitting S by variable i_0 . The event $\hat{S}^{ei} = S$ (correct recovery of S) requires the event $\frac{NC(M)}{NC(S)} < 1$ where NC is the normalized criterion in (9). That is S is preferred to model M .

$$P(\hat{S}^{ei} = S) \leq P\left(\frac{NC(M)}{NC(S)} < 1\right) = P\left(e^{C(M)-C(S)} < 1\right)$$

Using that $C(S) - C(M) = L_{SM}/2 + \Delta_{MS}$ for L_{SM} and Δ_{MS} defined in (S12), and that $\Delta_{MS} = -\kappa_j$ we obtain

$$P(\hat{S}^{ei} = S) \leq P\left(e^{-(\frac{1}{2}L_{SM} + \Delta_{MS})} < 1\right) = P\left(L_{SM} > 2\kappa_j\right). \quad (\text{S40})$$

Taking square root on both sides of the inequality in the right-most probability in (S40), we get

$$P(\hat{S}^{ei} = S) \leq P\left(\sqrt{L_{SM}} > \sqrt{2\kappa_j}\right). \quad (\text{S41})$$

By Lemma S3.7, $L_{SM} \in \chi_1^2(\mu_{SM})$ and there exists then $Z_M \sim N(\sqrt{\mu_{SM}}, 1)$ such that $\sqrt{L_{SM}} = |Z_M|$. We have

$$P(\hat{S}^{ei} = S) \leq P(|Z_M| > \sqrt{2\kappa_j}). \quad (\text{S42})$$

Consider first the case where $2\kappa_j > \mu_{SM}$. Using that $|Z_M| = |Z_M - \sqrt{\mu_{SM}} + \sqrt{\mu_{SM}}| \leq |Z_M - \sqrt{\mu_{SM}}| + \sqrt{\mu_{SM}}$, we get that

$$P(|Z_M| > \sqrt{2\kappa_j}) \leq P(|Z_M - \sqrt{\mu_{SM}}| > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) = 2P(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}})$$

where $z \sim N(0, 1)$ and the equality follows from the symmetry of the standard Gaussian distribution. Since $2\kappa_j > \mu_{SM}$, $P(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) < 1/2$, $P(|Z_M| > \sqrt{2\kappa_j}) < 1$ and $P(\hat{S}^{ei} = S) < 1$ too by (S42).

Consider next the case where $2\kappa_j \leq \mu_{SM}$. By Lemma S3.1, we have

$$P(|Z_M| > \sqrt{2\kappa_j}) = P(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) + P(z < -\sqrt{2\kappa_j} - \sqrt{\mu_{SM}})$$

where $z \sim N(0, 1)$. Since $\kappa_j \rightarrow +\infty$, $\sqrt{2\kappa_j} + \sqrt{\mu_{SM}} \rightarrow \infty$ and we have that $P(z < -\sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) \rightarrow 0$. Further, by Lemma S3.9 with $T = S$, $\sqrt{\mu_{SM}} \leq \sqrt{n\bar{\lambda}\beta_{\min,j}^*}$. By assumption, we have $\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\kappa_j} < \infty$. It follows that $\lim_{n \rightarrow \infty} \sqrt{\mu_{SM}} - \sqrt{2\kappa_j} < \infty$ and we have that $\lim_{n \rightarrow \infty} P(z > \sqrt{2\kappa_j} - \sqrt{\mu_{SM}}) < 1$. Hence, we get $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$, as claimed.

Part (iii). First, we re-write

$$\sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\lambda_j \ln(p_j - s_j)} = \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\kappa_j} + \sqrt{2\lambda_j \ln(p_j - s_j)} \left(\sqrt{\frac{\kappa_j}{\lambda_j \ln(p_j - s_j)}} - 1 \right).$$

By assumption $\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\lambda_j \ln(p_j - s_j)} < \infty$ and there exists $k \in \mathbb{R}^+$ such that

$$\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\kappa_j} + \sqrt{2\lambda_j \ln(p_j - s_j)} \left(\sqrt{\frac{\kappa_j}{\lambda_j \ln(p_j - s_j)}} - 1 \right) \leq k \quad (\text{S43})$$

Consider the case $\lim_{n \rightarrow \infty} \sqrt{\frac{\kappa_j}{\lambda_j \ln(p_j - s_j)}} - 1 < 0$. Since by assumption $\lim_{n \rightarrow +\infty} (p_j - s_j)^{\lambda_j} = +\infty$, $p_j - s_j$ diverges and then by Part (i), we have $\lim_{n \rightarrow \infty} \sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} = S) = 0$. Now consider the case $\lim_{n \rightarrow \infty} \sqrt{\frac{\kappa_j}{\lambda_j \ln(p_j - s_j)}} - 1 \geq 0$. Then, $\kappa_j \rightarrow \infty$ since by assumption $\lim_{n \rightarrow \infty} \lambda_j \ln(p_j - s_j) > 0$. Moreover, condition (S43) implies that $\lim_{n \rightarrow \infty} \sqrt{n\bar{\lambda}\beta_{\min,j}^*} - \sqrt{2\kappa_j} \leq k$. By Part (ii), there exists $\beta^* \in \mathfrak{B}$ such that $\lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) < 1$.

S3.2.6 Proof of Corollary 3.6

Since $\hat{S}^{sd}$ is $\hat{S}^{ei}$ with $b = 1$, the assumptions of Proposition 3.5 (iii) for $\hat{S}^{sd}$ are met and there exists $\beta^* \in \mathfrak{B}$ such that $\lim_{n \rightarrow \infty} P(\hat{S}^{sd} = S) < 1$. Since Assumptions A1 and A2 hold, by Theorem 3.3 we have that $\inf_{\beta^* \in \mathfrak{B}} \lim_{n \rightarrow \infty} P(\hat{S}^{ei} = S) = 1$.

S3.2.7 Proof of Theorem 3.7

The proof strategy is the same as that of Theorem 3.3, with suitable adjustments. The first step is to use Lemma S3.10 to bound $\mathbb{E}(NC(M))$ for every $M \notin \mathcal{T}(\kappa)$. The main difference is that in the proof of Theorem 3.3 we took $T = S$ in Lemma S3.10, whereas now we take a model $T = T^M \in \mathcal{T}(\kappa)$ that depends on M . Intuitively, T^M contains large truly non-zero parameters that are missed by M , and hence T^M should be chosen over M asymptotically. More precisely, we choose $T^M \in \mathcal{T}(\kappa)$ such that $T^M \setminus M \subseteq S^L(\kappa)$ and the elements in $M \setminus T^M$ are either inactive or in $S^S(\kappa)$. The latter condition and Assumption A1 ensure that the assumptions of Lemma S3.10 are met for any $\beta^* \in \mathfrak{B}$. We then get a bound $\mathbb{E}(NC(M)) \leq e^{-\psi A_{TM}}$ for every large enough n and any $\psi \in (0, 1)$, where $A_{TM} = \gamma \Delta_{MT^M} + \frac{1-\gamma}{6} \mu_{Q_{TM}M}$ (cf (S12) and (S7)), $\gamma \in (1/2, 1)$ is defined in (20) and $Q_{TM} = M \cup T^M$. The second step is to obtain a lower bound for A_{TM} , which gives an upper bound for $\mathbb{E}(NC(M))$ that holds for any $\beta^* \in \mathfrak{B}$ (distinct to that obtained in the proof of Theorem 3.3). The final step is to get an upper-bound $\sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M))$ that vanishes (as n grows) under the assumptions of Theorem 3.7 for any $\beta^* \in \mathfrak{B}$. Then Lemma S3.6 immediately implies that $\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \notin \mathcal{T}(\kappa))$ also vanishes.

For the first step of the proof, recall that the set $\mathcal{T}(\kappa)$ contains all the models that are a superset of $S^L(\kappa)$ and a subset of $S^L(\kappa) \cup S^I(\kappa)$. For any $M \notin \mathcal{T}(\kappa)$, take the unique $T^M \in \mathcal{T}(\kappa)$ such that $T^M = [M \cap S^I(\kappa)] \cup S^L(\kappa)$, which implies $M \cap S^I(\kappa) = T^M \cap S^I(\kappa)$. That is, T^M contains all the large signals plus the intermediate signals in M . To show that $\mathbb{E}(NC(M)) \leq e^{-\psi A_{TM}}$ we show that A_{TM} satisfies the conditions of Lemma S3.10, taking $T = T^M$. That is, we wish to show that three conditions hold: $A_{TM} > 0$, $|M \setminus T^M| = o(A_{TM})$, and $\mu_{Q_{TM}T^M} = o(A_{TM})$.

Observe that Δ_{MT^M} , defined in (S12), can be rewritten as $\Delta_{MT^M} = \sum_{j=1}^b (|M_j \setminus T_j^M| - |T_j^M \setminus M_j|) \kappa_j$ and then:

$$A_{TM} = \sum_{j=1}^b |M_j \setminus T_j^M| \gamma \kappa_j + \frac{1-\gamma}{6} \mu_{Q_{TM}M} - \sum_{j=1}^b |T_j^M \setminus M_j| \gamma \kappa_j$$

Since $\gamma < 1$, $(1 - \gamma)/6 > 0$, by Lemma S3.8, for every $n \in \mathbb{N}$ we have, for any $\beta^* \in \mathfrak{B}$,

$$A_{TM} \geq \sum_{j=1}^b |M_j \setminus T_j^M| \gamma \kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \min_{i \in T_j^M \setminus M_j} \beta_i^{*2} - \gamma \kappa_j \right). \quad (\text{S44})$$

We have that $T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))$ since $T^M \in \mathcal{T}(\kappa)$ and that $T^M \cap S^I(\kappa) = M \cap S^I(\kappa)$. It follows that $T^M \setminus M \subseteq S^L(\kappa)$. By definition of $S^L(\kappa)$, the rightmost component in (S44) is nonnegative and, if $|T^M \setminus M| \neq 0$, it is positive, for every n large enough. By Assumption A1, component $\gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j$ is nonnegative, and, if $|M \setminus T^M| \neq 0$, positive. Now, $M \neq T^M$ implies that necessarily $|M \setminus T^M| \neq 0$ or $|T^M \setminus M| \neq 0$ and then, for every n large enough, $A_{TM} > 0$, establishing the first condition required by Lemma S3.10. Regarding its second condition, if $|M \setminus T^M| = 0$, we immediately have $|M \setminus T^M| = o(A_{TM})$. If $|M \setminus T^M| \neq 0$, since the rightmost component in (S44) is nonnegative, we have

$$\frac{|M \setminus T^M|}{A_{TM}} \leq \left[\gamma \sum_{j=1}^b \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|} \kappa_j \right]^{-1} \leq \left[\gamma \min_{j=1, \dots, b} \kappa_j \right]^{-1}$$

where the last inequality follows from $\sum_{j=1}^b \frac{|M_j \setminus T_j^M|}{|M \setminus T^M|} = 1$. By Assumption A1, $\min_{j=1, \dots, b} \kappa_j \rightarrow \infty$ as $n \rightarrow \infty$ and hence $|M \setminus T^M| = o(A_{T^M})$ when $|M \setminus T^M| \neq 0$ too.

Finally, consider the third condition in Lemma S3.10. In the proof of Theorem 3.3, $\mu_{Q_S S} = o(A_S)$ is immediate because $Q_S \setminus S = M \setminus S \subseteq S^C$, i.e. since all parameters in $M \setminus S$ are truly zero we have that $\mu_{Q_S S} = 0$. Here $M \setminus T^M$ is not necessarily a subset of S^C , hence $\mu_{Q_{T^M} T^M} \geq 0$. Note that, since $T^M \in \mathcal{T}(\kappa)$, we have that $S^L(\kappa) \subseteq T^M$. Moreover, we have $M \cap S^I(\kappa) = T^M \cap S^I(\kappa)$. It follows that $M \setminus T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))^C$, that is the elements of $M \setminus T^M$ are either inactive or belong to $S^S(\kappa)$. If $M \cap S^S(\kappa) = \emptyset$ then $M \setminus T^M \subseteq S^C$ and we immediately get $\mu_{Q_{T^M} T^M} = o(A_{T^M})$ because $\beta_{M \setminus T^M}^* = \mu_{Q_{T^M} T^M} = 0$ for any $\beta^* \in \mathfrak{B}$. Assume now that $M \cap S^S(\kappa) \neq \emptyset$. Using that the rightmost component in (S44) is nonnegative and Lemma S3.9, we have

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{\mu_{Q_{T^M} T^M}}{\gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j}.$$

Observe that for all $j = 1, \dots, b$, $M_j \setminus T_j^M \supseteq (S_j \cap M_j) \setminus T_j^M$ and then $|M_j \setminus T_j^M| \geq |(S_j \cap M_j) \setminus T_j^M|$. We then get

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2}}{\gamma \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \kappa_j}.$$

Moreover, since $M \setminus T^M \subseteq (S^I(\kappa) \cup S^L(\kappa))^C$ as discussed earlier, we have, for all $j = 1, \dots, b$, $(S_j \cap M_j) \setminus T_j^M \subseteq S_j^S(\kappa_j)$. It follows $\max_{i \in (S_j \cap M_j) \setminus T_j^M} \beta_i^{*2} \leq \max_{i \in S_j^S} \beta_i^{*2}$ and

$$\frac{\mu_{Q_{T^M} T^M}}{A_{T^M}} \leq \frac{n \bar{\lambda} \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{\gamma \sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \kappa_j}. \quad (\text{S45})$$

Let $\bar{r} := \sum_{j=1}^b n \bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2} / (\gamma \kappa_j)$. We show next that $\bar{r}$ is an upper bound on $\mu_{Q_{T^M} T^M} / A_{T^M}$. By restricting the sum in $\bar{r}$ to the j such that $|(S_j \cap M_j) \setminus T_j^M| \neq 0$ and multiplying the numerator and denominator of the summand by $|(S_j \cap M_j) \setminus T_j^M|$, we get

$$\bar{r} \geq \sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b \frac{|(S_j \cap M_j) \setminus T_j^M| n \bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{|(S_j \cap M_j) \setminus T_j^M| \gamma \kappa_j} \quad (\text{S46})$$

Note that for any collections $(\alpha_j, \delta_j) \in \mathbb{R} \times \mathbb{R} \setminus \{0\}$, $j = 1, \dots, b$, we have

$$\sum_{j=1}^b \frac{\alpha_j}{\delta_j} = \sum_{j=1}^b \frac{\alpha_j \frac{1}{\delta_j} (\delta_j + \sum_{l \neq j} \delta_l)}{\sum_{j=1}^b \delta_j} = \frac{\sum_{j=1}^b \alpha_j (1 + \sum_{l \neq j} \frac{\delta_l}{\delta_j})}{\sum_{j=1}^b \delta_j} \quad (\text{S47})$$

Using (S47) in the right-hand side of (S46), we get

$$\begin{aligned} \bar{r} &\geq \frac{\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| n \bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2} \left(1 + \sum_{l \neq j} \frac{|(S_l \cap M_l) \setminus T_l^M| \gamma \kappa_l}{|(S_j \cap M_j) \setminus T_j^M| \gamma \kappa_j}\right)}{\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| \gamma \kappa_j} \\ &\geq \frac{\sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| n \bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{\sum_{j=1}^b |(S_j \cap M_j) \setminus T_j^M| \gamma \kappa_j} \end{aligned}$$

where last inequality follows from $\left(1 + \sum_{l \neq j} \frac{|(S_l \cap M_l) \setminus T_l^M| \gamma \kappa_l}{|(S_j \cap M_j) \setminus T_j^M| \gamma \kappa_j}\right) \geq 1$ for all j and from the identity $\sum_{j=1, |(S_j \cap M_j) \setminus T_j^M| \neq 0}^b |(S_j \cap M_j) \setminus T_j^M| \gamma \kappa_j = \sum_{j=1}^b |(S \cap M_j) \setminus T_j^M| \gamma \kappa_j$. Then, by (S45),

$$\frac{\mu_{Q_{TM}TM}}{A_{TM}} \leq \bar{r} = \sum_{j=1}^b \frac{n\bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2}}{\gamma \kappa_j}.$$

By definition of $S_j^S(\kappa_j)$, $n\bar{\lambda} \max_{i \in S_j^S(\kappa_j)} \beta_i^{*2} = o(\kappa_j)$ for all j and $\mu_{Q_{TM}TM} = o(A_{TM})$ for any $\beta^* \in \mathfrak{B}$ when $M \cap S^S(\kappa) \neq \emptyset$ too. We can now apply Lemma S3.10 and get that for every $M \in \mathcal{M} \setminus \mathcal{T}(\kappa)$, for any $\beta^* \in \mathfrak{B}$ and $\psi \in (0, 1)$, and for every n large enough, $\mathbb{E}(NC(M)) \leq e^{-\psi A_{TM}}$.

The second step of the proof is to lower-bound $A_{TM} = \gamma \Delta_{MTM} + \frac{1-\gamma}{6} \mu_{Q_{TM}TM}$. Let

$$A_{TM}^* := \gamma \sum_{j=1}^b |M_j \setminus T_j^M| \kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma \kappa_j \right)$$

Recall that, for every j , $T_j^M \setminus M_j \subseteq S_j^L(\kappa_j)$. We have then $\min_{i \in T_j^M \setminus M_j} \beta_i^{*2} \geq \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2}$, and by (S44), $A_{TM} \geq A_{TM}^*$. It follows that for any $\psi \in (0, 1)$ and for any $\beta^* \in \mathfrak{B}$ and every n large enough,

$$\mathbb{E}(NC(M)) \leq e^{-\psi A_{TM}^*}. \quad (\text{S48})$$

To conclude the second part of the proof we lower-bound $\psi A_{TM}^* = \sum_{j=1}^b |M_j \setminus T_j^M| \psi \gamma \kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma \kappa_j \right)$. To do this, we obtain a lower bound for $\psi \gamma \kappa_j$ and for $\psi \left(\frac{1-\gamma}{6} n \rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma \kappa_j \right)$.

The definition of $S_j^L(\kappa_j)$ implies there exists some $g'_j \rightarrow \infty$ such that

$$\frac{(1-\gamma)n\rho(\mathbf{X})}{6} \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \kappa_j = \ln(s_j) + g'_j. \quad (\text{S49})$$

Let $\delta \in (0, 1)$ and denote $\bar{m}_j = \max \left\{ \frac{2 \ln(p_j - s_j)}{f_j}, \frac{2 \ln(s_j)}{g'_j} \right\}$, where f_j is given in Assumption A1.

Take $\psi = \max_{j=1, \dots, b} \frac{\xi + \delta + \bar{m}_j}{1 + \bar{m}_j}$ for some $\xi \in (0, 1 - \delta)$ then $\psi \in (0, 1)$, and we have, for every $j = 1, \dots, b$,

$$\psi > \frac{\delta + \frac{2 \ln(p_j - s_j)}{f_j}}{1 + \frac{2 \ln(p_j - s_j)}{f_j}} = \frac{\delta f_j / 2 + \ln(p_j - s_j)}{f_j / 2 + \ln(p_j - s_j)} \quad (\text{S50})$$

$$\psi > \frac{\delta + \frac{2 \ln(s_j)}{g'_j}}{1 + \frac{2 \ln(s_j)}{g'_j}} = \frac{\delta g'_j / 2 + \ln(s_j)}{g'_j / 2 + \ln(s_j)} \geq \frac{\delta g'_j / 2 + \ln(s_j)}{g'_j + \ln(s_j)}. \quad (\text{S51})$$

Recall that Assumptions A1 defines $f_j = \kappa_j - \ln(p_j - s_j)$ and $\gamma = \frac{1}{2}(1 + \max_j \frac{\ln(p_j - s_j)}{\kappa_j})$ respectively. Hence,

$$\gamma \kappa_j \geq \frac{1}{2} \left(1 + \frac{\ln(p_j - s_j)}{\kappa_j} \right) \kappa_j = \ln(p_j - s_j) + \frac{1}{2} (\kappa_j - \ln(p_j - s_j)) = \ln(p_j - s_j) + \frac{1}{2} f_j.$$

Hence, by (S50), we have

$$\psi\gamma\kappa_j \geq \psi\left(\ln(p_j - s_j) + \frac{1}{2}f_j\right) \geq \ln(p_j - s_j) + \delta\frac{1}{2}f_j. \quad (\text{S52})$$

Further,

$$\psi\left(\frac{1-\gamma}{6}n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma\kappa_j\right) \geq \psi\left(\ln(s_j) + g'_j\right) \geq \ln(s_j) + \delta\frac{1}{2}g'_j. \quad (\text{S53})$$

where the first inequality follows from (S49) and the second inequality from (S51). In (S48), $\psi A_{TM}^* = \sum_{j=1}^b |M_j \setminus T_j^M| \psi\gamma\kappa_j + \sum_{j=1}^b |T_j^M \setminus M_j| \psi\left(\frac{1-\gamma}{6}n\rho(\mathbf{X}) \min_{i \in S_j^L(\kappa_j)} \beta_i^{*2} - \gamma\kappa_j\right)$. Then by (S52) and (S53), we get, for any $\beta^* \in \mathfrak{B}$,

$$\mathbb{E}(NC(M)) \leq \exp\left\{-\sum_{j=1}^b |M_j \setminus T_j^M| (\ln(p_j - s_j) + \delta\frac{f_j}{2}) - \sum_{j=1}^b |T_j^M \setminus M_j| (\ln(s_j) + \delta\frac{g'_j}{2})\right\}. \quad (\text{S54})$$

For the final step of the proof, denote $\mathcal{S} = \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M))$ for convenience. Since (S54) holds for any $\beta^* \in \mathfrak{B}$, we have

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} e^{-\sum_{j=1}^b |M_j \setminus T_j^M| \left(\ln(p_j - s_j) + \delta\frac{f_j}{2}\right) - \sum_{j=1}^b |T_j^M \setminus M_j| \left(\ln(s_j) + \delta\frac{g'_j}{2}\right)}.$$

We split the sum in the right-hand side above into sums over models M such that $T^M = T$ for some common $T \in \mathcal{T}(\kappa)$. Denote for any $T \in \mathcal{T}$, $\mathcal{M}(T) := \{M \in \mathcal{M} \setminus \mathcal{T}(\kappa) \mid T^M = T\}$, then

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq \sum_{T \in \mathcal{T}(\kappa)} \sum_{M \in \mathcal{M}(T)} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta\frac{f_j}{2}\right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta\frac{g'_j}{2}\right)}. \quad (\text{S55})$$

The right hand-side of (S55) is composed of a double sum over $T \in \mathcal{T}(\kappa)$ and over $M \in \mathcal{M}(T)$. Consider the sum over $M \in \mathcal{M}(T)$, add T to it, and denote it

$$\mathcal{S}(T) = \sum_{M \in \mathcal{M}(T) \cup T} e^{-\sum_{j=1}^b |M_j \setminus T_j| \left(\ln(p_j - s_j) + \delta\frac{f_j}{2}\right) - \sum_{j=1}^b |T_j \setminus M_j| \left(\ln(s_j) + \delta\frac{g'_j}{2}\right)}. \quad (\text{S56})$$

In the summand in the right-hand side of (S56), the case $M = T$ correspond to $|M_j \setminus T_j^M| = |T_j^M \setminus M_j| = 0$ for all j and the summand is then 1. By (S55), we then get that

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq \sum_{T \in \mathcal{T}(\kappa)} (\mathcal{S}(T) - 1). \quad (\text{S57})$$

For each $T = T^M$, we further split $\mathcal{S}(T)$ into sums over subsets of models M that have u_j more parameters than T^M in block j , and are missing w_j parameters from T^M in block j . Specifically, consider models M such that, for all j , $|M_j \setminus T_j^M| = u_j$ and $|T_j^M \setminus M_j| = w_j$ with $u_j \in \{0, \dots, p_j - s_j, \dots, p_j - s_j + |S_j^S(\kappa_j)|\}$ and $w_j \in \{0, \dots, |S_j^L(\kappa_j)|\}$. Denote by

$$\mathcal{S}_{\mathbf{w}}^{\mathbf{u}}(T) = \sum_{M \in \mathcal{M}(T) \cup T: \forall j, |M_j \setminus S_j| = u_j, |S_j \setminus M_j| = w_j} e^{-\sum_{j=1}^b u_j \left(\ln(p_j - s_j) + \delta\frac{f_j}{2}\right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta\frac{g'_j}{2}\right)}.$$

We get

$$\mathcal{S}(T) = \sum_{w_1=0}^{|S_1^L(\kappa)|} \cdots \sum_{w_b=0}^{|S_b^L(\kappa)|} \sum_{u_1=0}^{p_1-s_1+|S_1^S|} \cdots \sum_{u_b=0}^{p_b-s_b+|S_b^S|} \mathcal{S}_{\mathbf{w}}^{\mathbf{u}}(T). \quad (\text{S58})$$

The number of models missing, for all j , w_j out of the $|S_j^L(\kappa)|$ large active parameters and having u_j inactive or small active parameters from B_j is $\prod_{j=1}^b \binom{p_j-s_j+|S_j^S(\kappa)|}{u_j} \binom{|S_j^L(\kappa)|}{w_j}$. We thus have

$$\begin{aligned} \mathcal{S}_{\mathbf{w}}^{\mathbf{u}}(T) &= \left(\prod_{j=1}^b \binom{p_j-s_j+|S_j^S(\kappa)|}{u_j} \binom{|S_j^L(\kappa)|}{w_j} \right) e^{-\sum_{j=1}^b u_j \left(\ln(p_j-s_j) + \delta \frac{f_j}{2} \right) - \sum_{j=1}^b w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \\ &= \prod_{j=1}^b \binom{p_j-s_j+|S_j^S(\kappa)|}{u_j} e^{-u_j \left(\ln(p_j-s_j) + \delta \frac{f_j}{2} \right)} \binom{|S_j^L(\kappa)|}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)}. \end{aligned}$$

Inputting the expression above in (S58) and factorizing over terms in u_j and w_j gives

$$\begin{aligned} \mathcal{S}(T) &\leq \prod_{j=1}^b \left(\sum_{u_j=0}^{p_j-s_j+|S_j^S(\kappa)|} \binom{p_j-s_j+|S_j^S(\kappa)|}{u_j} e^{-u_j \left(\ln(p_j-s_j) + \delta \frac{f_j}{2} \right)} \right) \\ &\quad \cdot \left(\sum_{w_j=0}^{|S_j^L(\kappa)|} \binom{|S_j^L(\kappa)|}{w_j} e^{-w_j \left(\ln(s_j) + \delta \frac{g'_j}{2} \right)} \right). \end{aligned}$$

By the bound in (S25) and taking the terms in $u_j = 0$ and $w_j = 0$ out of the sums above, we have

$$\begin{aligned} \mathcal{S}(T) &\leq \prod_{j=1}^b \left(1 + \sum_{u_j=1}^{p_j-s_j+|S_j^S(\kappa_j)|} e^{-u_j \left(\delta \frac{f_j}{2} - \ln \left(1 + \frac{|S_j^S(\kappa_j)|}{p_j-s_j} \right) - 1 \right)} \right) \\ &\quad \cdot \left(1 + \sum_{w_j=1}^{|S_j^L(\kappa_j)|} e^{-w_j \left(\delta \frac{g'_j}{2} + \ln \left(\frac{s_j}{|S_j^L(\kappa_j)|} \right) - 1 \right)} \right). \end{aligned} \quad (\text{S59})$$

Denote

$$d_j = e^{1 + \ln \left(1 + \frac{|S_j^S(\kappa_j)|}{p_j-s_j} \right) - \delta \frac{f_j}{2}}, \quad h_j = e^{1 - \ln \left(\frac{s_j}{|S_j^L(\kappa_j)|} \right) - \delta \frac{g'_j}{2}}.$$

By assumption $|S_j^S(\kappa_j)| = O(p_j - s_j)$, and by definition of $|S_j^L(\kappa_j)|$ we have $s_j \geq |S_j^L(\kappa_j)|$. By Assumptions A1 and the definition of $|S_j^L(\kappa_j)|$, we also have $f_j \rightarrow \infty$ and $g'_j \rightarrow \infty$, and then $\lim_{n \rightarrow \infty} d_j = \lim_{n \rightarrow \infty} h_j = 0$. Using the properties of geometric series, for every j we have

$$\begin{aligned} 1 + \sum_{u_j=1}^{p_j-s_j+|S_j^S(\kappa_j)|} e^{-u_j \left(\delta \frac{f_j}{2} - \ln \left(1 + \frac{|S_j^S(\kappa_j)|}{p_j-s_j} \right) - 1 \right)} &= \frac{1 - d_j^{p_j-s_j+|S_j^S(\kappa_j)|+1}}{1-d_j} \\ 1 + \sum_{w_j=1}^{|S_j^L(\kappa_j)|} e^{-w_j \left(\delta \frac{g'_j}{2} + \ln \left(\frac{s_j}{|S_j^L(\kappa_j)|} \right) - 1 \right)} &= \frac{1 - h_j^{|S_j^L(\kappa_j)|+1}}{1-h_j}, \end{aligned}$$

where both expressions converge to 1 as n grows. By (S57) and (S59):

$$\sup_{\beta^* \in \mathfrak{B}} \mathcal{S} \leq \sum_{T \in \mathcal{T}} \left(\prod_{j=1}^b \left(\frac{1 - d_j^{p_j-s_j+|S_j^S(\kappa_j)|+1}}{1-d_j} \right) \left(\frac{1 - h_j^{|S_j^L(\kappa_j)|+1}}{1-h_j} \right) - 1 \right).$$

Each of the summand vanishes as $n \rightarrow \infty$. Moreover, by assumption $|S^I(\kappa)| = O(1)$ and then $|\mathcal{T}(\kappa)| = 2^{|S^I(\kappa)|} = O(1)$. We thus have $\lim_{n \rightarrow \infty} \sup_{\beta^* \in \mathfrak{B}} \mathcal{S} = \lim_{n \rightarrow \infty} \sup_{\beta^* \in \mathfrak{B}} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 0$.

Further, by Lemma S3.6, $\sup_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \notin \mathcal{T}(\kappa)) \leq (|\mathcal{T}(\kappa)| + 1) \sup_{\beta^* \in \mathfrak{B}} \mathcal{S} = (2^{|S^I(\kappa)|} + 1) \sup_{\beta^* \in \mathfrak{B}} \mathcal{S}$. Since $\sup_{\beta^* \in \mathfrak{B}} \mathcal{S}$ vanishes and $|S^I(\kappa)| = O(1)$, $\lim_{n \rightarrow \infty} \inf_{\beta^* \in \mathfrak{B}} P(\hat{S}^{ei} \in \mathcal{T}(\kappa)) = 1$ as claimed.

S3.3 Proofs of Section 4

S3.3.1 Proof of Proposition 4.2

Denote

$$A_j := \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{T}(\kappa) | i \in M} NC(M) \quad \text{and} \quad C_j := \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa) | i \in M} NC(M).$$

For every $j = 1, \dots, b$, we have the decomposition

$$\frac{\hat{s}_j}{p_j} = \frac{\sum_{i \in B_j} \sum_{M \in \mathcal{M} | i \in M} NC(M)}{p_j} = A_j + C_j. \quad (\text{S60})$$

To show the lower bound on $\hat{s}_j/p_j$, we decompose A_j

$$\begin{aligned} A_j &= \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j} \frac{I(i \in M_j)}{p_j} \\ &= \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in S_j^L(\kappa_j)} \frac{I(i \in M_j)}{p_j} + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j \setminus S_j^L(\kappa_j)} \frac{I(i \in M_j)}{p_j} \\ &= \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j \setminus S_j^L(\kappa_j)} \frac{I(i \in M_j)}{p_j}. \end{aligned} \quad (\text{S61})$$

where the last equality follows from $I(i \in M_j) = 1$ for all $i \in S^L(\kappa)$ when $M \in \mathcal{T}(\kappa)$. The rightmost term above and C_j are nonnegative, then by the linearity of the expectation

$$\mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \geq \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)).$$

By Theorem 3.7, $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 1$. It follows that $\lim_{n \rightarrow \infty} \mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \geq \frac{|S_j^L(\kappa_j)|}{p_j}$ for every $j = 1, \dots, b$.

We now prove the upper bound. Recall that $\mathcal{T}(\kappa)$ by definition includes models that have no small signals, i.e. all parameters are in $S^L(\kappa) \cup S^I(\kappa)$. That is, for all $M \in \mathcal{T}(\kappa)$, we have that $I(i \in M_j) = 0$ for all $i \in B_j \setminus (S_j^L(\kappa_j) \cup S_j^I(\kappa_j))$. Hence, A_j in (S61) satisfies

$$\begin{aligned} A_j &= \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \sum_{M \in \mathcal{T}(\kappa)} NC(M) \sum_{i \in S_j^I(\kappa_j)} \frac{I(i \in M_j)}{p_j} \\ &\leq \frac{|S_j^L(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + \frac{|S_j^I(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) \end{aligned}$$

where the inequality follows from $\sum_{i \in S_j^I(\kappa_j)} I(i \in M_j) \leq |S_j^I(\kappa_j)|$ for all M . By (S60), we then have

$$\frac{\hat{s}_j}{p_j} \leq \frac{|S_j^L(\kappa_j)| + |S_j^I(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} NC(M) + C_j. \quad (\text{S62})$$

Moreover, for every $j = 1, \dots, b$, C_j satisfies

$$C_j = \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} NC(M) \sum_{i \in B_j} \frac{I(i \in M_j)}{p_j} \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} NC(M) \quad (\text{S63})$$

where the inequality follows from $\sum_{i \in B_j} \frac{I(i \in M_j)}{p_j} \leq 1$ for all M . Taking expectations in (S62) and (S63) gives

$$\mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \leq \frac{|S_j^L(\kappa_j)| + |S_j^I(\kappa_j)|}{p_j} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) + \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M)).$$

By Theorem 3.7, we have on one hand $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 1$ and, on the other hand, $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa)} \mathbb{E}(NC(M)) = 0$. It follows that $\lim_{n \rightarrow \infty} \mathbb{E}\left(\frac{\hat{s}_j}{p_j}\right) \leq \frac{|S_j^L(\kappa_j)| + |S_j^I(\kappa_j)|}{p_j} = \frac{s_j - |S_j^S(\kappa_j)|}{p_j}$ for every $j = 1, \dots, b$, which proves the upper bound.

S3.3.2 Proof of Lemma 4.1

Denote the marginal likelihood of $\mathbf{y}$ given $\boldsymbol{\theta}$ by $H(\boldsymbol{\theta}) = p(\mathbf{y} \mid \boldsymbol{\theta})$. For every $j = 1, \dots, b$, the partial derivative of its logarithm with respect to θ_j is

$$\frac{\partial \ln H(\boldsymbol{\theta})}{\partial \theta_j} = \frac{\partial H(\boldsymbol{\theta})}{\partial \theta_j} H(\boldsymbol{\theta})^{-1}. \quad (\text{S64})$$

Observe that:

$$H(\boldsymbol{\theta}) = \sum_{M \in \mathcal{M}} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \quad \text{and} \quad \frac{\partial H(\boldsymbol{\theta})}{\partial \theta_j} = \sum_{M \in \mathcal{M}} p(\mathbf{y} \mid M, \boldsymbol{\theta}) \frac{\partial p(M \mid \boldsymbol{\theta})}{\partial \theta_j} \quad (\text{S65})$$

Recall that each model is defined as $M = (m_1, \dots, m_p)$ where $m_i = I(\beta_i \neq 0)$ indicates whether variable j is included under M , and that our choice of model prior in (6) is

$$p(M \mid \boldsymbol{\theta}) = \prod_{j=1}^b (\theta_j)^{\sum_{i \in B_j} m_i} (1 - \theta_j)^{p_j - \sum_{i \in B_j} m_i}.$$

Hence, simple algebra shows that for every $M \in \mathcal{M}$

$$\frac{\partial p(M \mid \boldsymbol{\theta})}{\partial \theta_j} = p(M \mid \boldsymbol{\theta}) \left(\frac{\sum_{i \in B_j} m_i}{\theta_j} - \frac{p_j - \sum_{i \in B_j} m_i}{1 - \theta_j} \right). \quad (\text{S66})$$

Replacing (S66) into (S65), and using that for any function f

$$\sum_{M \in \mathcal{M}} \sum_{i \in B_j} m_i f(M) = \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i=1} f(M),$$

we get

$$\begin{aligned} \frac{\partial H(\boldsymbol{\theta})}{\partial \theta_j} &= \frac{1}{\theta_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \\ &\quad - \frac{1}{1 - \theta_j} \left[p_j H(\boldsymbol{\theta}) - \sum_{i \in B_j} \sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta}) \right] \end{aligned} \quad (\text{S67})$$

Note that

$$\frac{\sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y} \mid M, \boldsymbol{\theta}) p(M \mid \boldsymbol{\theta})}{H(\boldsymbol{\theta})} = \frac{\sum_{M \in \mathcal{M}: m_i=1} p(\mathbf{y}, M \mid \boldsymbol{\theta})}{p(\mathbf{y} \mid \boldsymbol{\theta})} = P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta})$$

By (S64), we then get the following expression of the partial derivative, for every $j = 1, \dots, b$,

$$\frac{\partial \ln H(\boldsymbol{\theta})}{\partial \theta_j} = \frac{1}{\theta_j} \sum_{i \in B_j} P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) - \frac{1}{1 - \theta_j} \left[p_j - \sum_{i \in B_j} P(m_i = 1 \mid \mathbf{y}, \boldsymbol{\theta}) \right]$$

Setting the partial derivatives to 0 and solving for θ_j gives the desired result.

S3.3.3 Proof of Theorem 4.3

The proof strategy is to show Assumptions A1 and A2 hold to apply Theorem 3.3. We first derive a convenient decomposition of the penalties κ_j^A . The second step is to show that, under Assumption A3, κ_j^A satisfies Assumption A1 with probability going to 1 as n grows. The third step is to show that Assumption A4 implies Assumption A2 for the κ_j^A with probability going to 1. The consistency of $\hat{S}^{A,ei}$ then follows from Theorem 3.3.

Denote for any $M \in \mathcal{M}$, $NC^\circ(M)$, the normalized criterion value for model M under Step 1 penalty κ° . For this choice of penalty and every $j = 1, \dots, b$, we have

$$\begin{aligned} \frac{\hat{s}_j}{p_j} &= \frac{1}{p_j} \sum_{i \in B_j} \sum_{M \in \mathcal{M}} NC^\circ(M) I(i \in M) \\ &= \frac{1}{p_j} \sum_{M \in \mathcal{M}} NC^\circ(M) \sum_{i \in B_j} I(i \in M) \\ &= \frac{s_j}{p_j} NC^\circ(S) + \sum_{M \in \mathcal{M} \mid M \neq S} \frac{|M_j|}{p_j} NC^\circ(M). \end{aligned}$$

Using that $NC^\circ(S) = 1 - \sum_{M \in \mathcal{M} \mid M \neq S} NC^\circ(M)$, we get

$$\frac{\hat{s}_j}{p_j} = \frac{s_j}{p_j} + \sum_{M \in \mathcal{M} \mid M \neq S} \frac{|M_j| - s_j}{p_j} NC^\circ(M).$$

Consider the decomposition of the sum in the right-hand side above between the sum over models M that contain more parameters than S in block j and the sum over those that contain fewer parameters than S in block b . Denote

$$O_j^\circ := \sum_{M \in \mathcal{M} \mid |M_j| > s_j} \frac{|M_j| - s_j}{p_j} NC^\circ(M) \quad \text{and} \quad U_j^\circ := \sum_{M \in \mathcal{M} \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M).$$

We have

$$\frac{\hat{s}_j}{p_j} = \frac{s_j}{p_j} + O_j^\circ - U_j^\circ. \quad (\text{S68})$$

Observe that we have the following decomposition of Step 2 penalties

$$\kappa_j^A = \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(\frac{p_j - \hat{s}_j}{p_j - s_j}\right).$$

By (S68), it follows that

$$\kappa_j^A = \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 - \frac{p_j(O_j^\circ - U_j^\circ)}{p_j - s_j}\right), \quad (\text{S69})$$

completing the first step of the proof.

We continue with the second step of the proof: showing that the κ_j^A 's satisfy Assumption A1 with probability going to 1. Recall that Assumption A1 states that there exists $f_j \rightarrow \infty$ (as $n \rightarrow \infty$) such that for every sufficiently large n ,

$$\kappa_j = \ln(p_j - s_j) + f_j.$$

Since U_j° is nonnegative, a lower bound on κ_j^A is

$$\kappa_j^A \geq \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right). \quad (\text{S70})$$

Plugging in the definition of O_j° , we have that

$$\frac{p_j O_j^\circ}{p_j - s_j} = \sum_{M \in \mathcal{M} \mid |M_j| > s_j} \frac{|M_j| - s_j}{p_j - s_j} NC^\circ(M) \leq \sum_{M \in \mathcal{M} \mid |M_j| > s_j} NC^\circ(M)$$

where the inequality follows from $(|M_j| - s_j)/(p_j - s_j) \leq 1$ for all M . Note that if M is such that $|M_j| > s_j$, then $M \notin \mathcal{T}(\kappa^\circ)$ (this follows immediately from the definition of $\mathcal{T}(\kappa)$ in (21)) and therefore $\sum_{M \in \mathcal{M} \mid |M_j| > s_j} NC^\circ(M) \leq \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} NC^\circ(M)$. Moreover, κ° satisfies Assumption A1, Assumption A3 is assumed to hold and the assumptions of Theorem 3.7 are met for κ° . Then, by Theorem 3.7, $\lim_{n \rightarrow \infty} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ)} \mathbb{E}(NC^\circ(M)) = 0$. It follows that $\frac{p_j O_j^\circ}{p_j - s_j}$ vanishes in probability and so does $\ln\left(1 - \frac{p_j O_j^\circ}{p_j - s_j}\right)$.

With probability going to 1, we then have that

$$\kappa_j^A \geq \ln(p_j - s_j) + \ln(\sqrt{n}) \quad (\text{S71})$$

and hence that the κ_j^A 's satisfy Assumption A1.

For the third part of the proof, we now show that Assumption A4 implies Assumption A2 for the κ_j^A . Assumption A2 for the κ_j^A states that for each block j there exists $g_j \rightarrow \infty$ such that for large enough n ,

$$\sqrt{\frac{(1 - \gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* - \sqrt{\kappa_j^A} = \sqrt{\ln(s_j)} + g_j.$$

where γ takes value

$$\gamma = \frac{1}{2} \left(1 + \max_j \frac{\ln(p_j - s_j)}{\kappa_j^A} \right). \quad (\text{S72})$$

To show that Assumption A4 implies Assumption A2 for the κ_j^A with probability going to 1, it suffices to show that the following two inequalities

$$\sqrt{\frac{(1-\gamma)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^* \geq \sqrt{\frac{(1-\xi)n\rho(\mathbf{X})}{6}} \beta_{\min,j}^*, \quad \text{and} \quad (\text{S73})$$

$$-\sqrt{\kappa_j^A} \geq -\sqrt{\ln\left(p - |S^L(\kappa^\circ)|\right) + \frac{1}{2}\ln(n)} \quad (\text{S74})$$

hold with probability going to 1 for γ as in (S72) and $\xi = \frac{1}{2} \left(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + 0.5 \ln(n)} \right)$ (defined in Assumption A4). We first show (S73) holds with probability going to 1 and then that (S74) does too.

By (S71) we have that with probability going to 1, for any j

$$\frac{\ln(p_j - s_j)}{\kappa_j^A} \leq \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + \ln(n)/2}.$$

It follows that

$$\gamma = \frac{1}{2} \left(1 + \max_{j=1,\dots,b} \frac{\ln(p_j - s_j)}{\kappa_j^A} \right) \leq \frac{1}{2} \left(1 + \max_j \frac{\ln(p_j - s_j)}{\ln(p_j - s_j) + \ln(n)/2} \right) = \xi$$

and (S73) holds with probability going to 1.

We now upper bound κ_j^A to show (S74) holds with probability going to 1. By (S69), we can write

$$\kappa_j^A = \kappa_j^{EB} + \ln(\hat{s}_j) = \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 - \frac{p_j(O_j^\circ - U_j^\circ)}{p_j - s_j}\right).$$

Since $O_j^\circ \geq 0$, we obtain that

$$\kappa_j^A \leq \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 + \frac{p_j}{p_j - s_j} U_j^\circ\right).$$

We split the sum in U_j° between models in $\mathcal{T}(\kappa^\circ)$ and those not in $\mathcal{T}(\kappa^\circ)$.

$$U_j^\circ = \sum_{M \in \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M) + \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} \frac{s_j - |M_j|}{p_j} NC^\circ(M).$$

If $M \in \mathcal{T}(\kappa^\circ)$, then by definition $|M_j| \geq |S_j^L(\kappa^\circ)|$ and thus $s_j - |M_j| \leq s_j - |S_j^L(\kappa^\circ)|$. A bound on $s_j - |M_j|$ for $M \notin \mathcal{T}(\kappa^\circ)$ is simply $s_j - |M_j| \leq s_j$. It follows that

$$U_j^\circ \leq \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j} \sum_{M \in \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M) + \frac{s_j}{p_j} \sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M)$$

By Theorem 3.7, $\sum_{M \in \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M)$ and $\sum_{M \in \mathcal{M} \setminus \mathcal{T}(\kappa^\circ) \mid |M_j| < s_j} NC^\circ(M)$ converge in probability to 1 and 0 respectively. We then get that, with probability going to 1,

$$\frac{p_j}{p_j - s_j} U_j^\circ \leq \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j - s_j} \quad (\text{S75})$$

By (S75), with probability going to 1,

$$\begin{aligned} \kappa_j^A &\leq \ln(p_j - s_j) + \ln(\sqrt{n}) + \ln\left(1 + \frac{s_j - |S_j^L(\kappa^\circ)|}{p_j - s_j}\right) \\ &= \ln(p_j - |S_j^L(\kappa^\circ)|) + \frac{1}{2} \ln(n). \end{aligned} \quad (\text{S76})$$

which shows (S74) holds with probability going to 1 and that Assumption A4 implies Assumption A2 holds for the κ_j^A with probability going to 1.

Since Assumptions A1 and A2 hold with probability going to 1, by Theorem 3.3, $\inf_{\beta^* \in \mathfrak{B}} \lim_{n \rightarrow \infty} P(\hat{S}^{A,ei} = S) = 1$, as claimed.

S3.4 Proof of Corollary 5.1

The result follows directly from Theorem 4.3, given that κ° is assumed to satisfy Assumption A3 and that Assumption A4 holds under Assumption A5.

References

- Birgé, L. (2001), ‘An alternative point of view on Lepski’s method’, *Lecture Notes-Monograph Series* **36**, 113–133.
- Hartigan, J. A. (2014), ‘Bounding the maximum of dependent random variables’, *Electronic Journal of Statistics* **8**(2), 3126 – 3140.
- Rossell, D. (2022), ‘Concentration of posterior model probabilities and normalized L_0 criteria’, *Bayesian Analysis* **17**(2), 565 – 591.

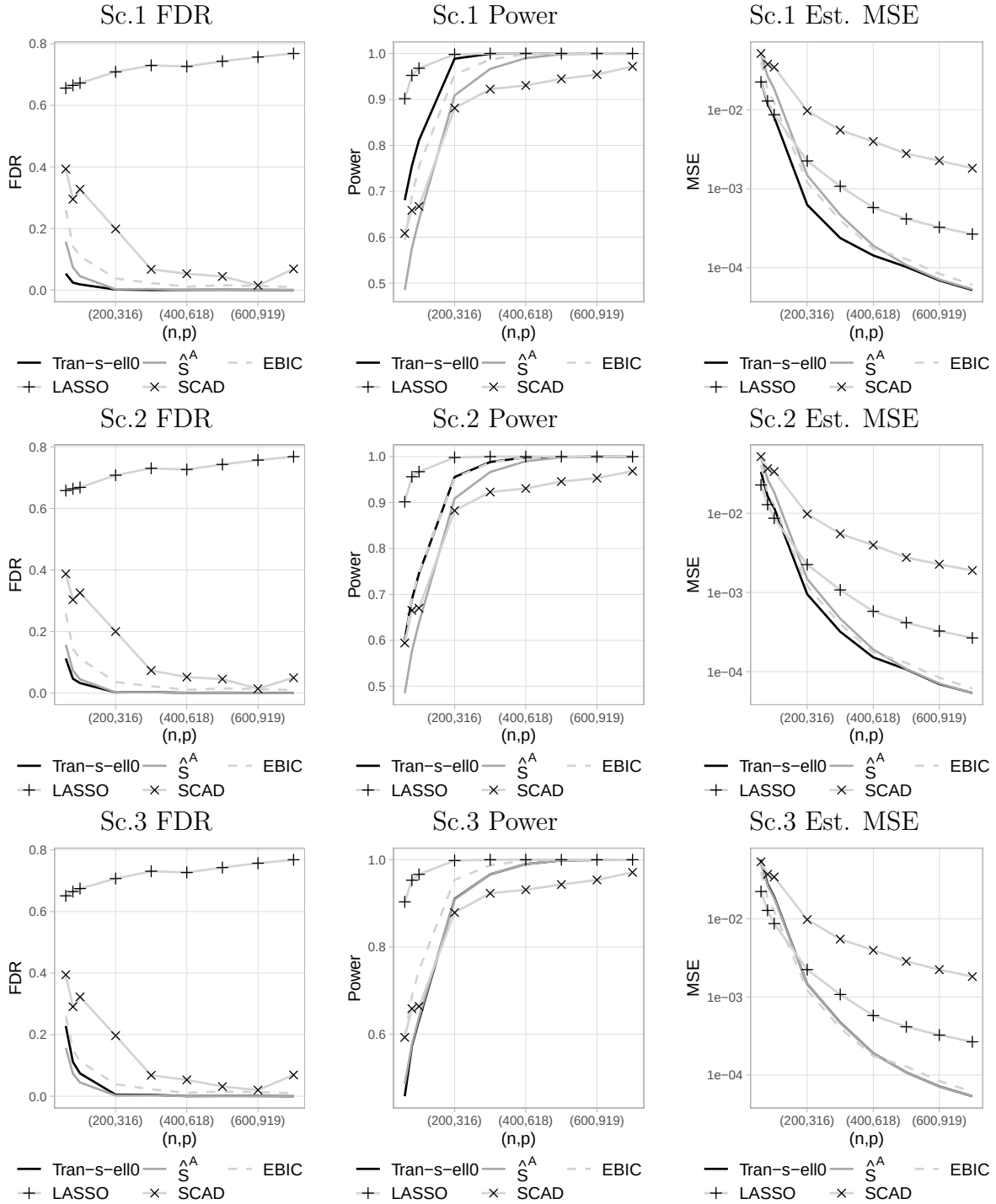


Figure S5: FDR (left), power (middle) and estimation MSE (right) with Tran-s-ell0, $\hat{S}^{A, sd}$, EBIC, LASSO and SCAD in Scenario 1 (top), 2 (middle) and 3 (bottom) as in Section 6.2